

The Chevalley-Weil Formula

On the Galois Module Structure of the
Space of Regular Differentials of a Curve
in Case it is Semi-Simple

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Abstract

We discuss Galois coverings X/Y of complete, nonsingular algebraic curves over an algebraically closed field. The Galois group $\text{Cov}(X/Y)$, which we denote by G , acts in a natural way on the space of regular differentials $H^0(X, \Omega_X^1)$. In terms of representation theory, we are given a linear representation

$$G \rightarrow \text{GL}(H^0(X, \Omega_X^1)).$$

The main question in this thesis is how many times a given irreducible representation of G occurs in $H^0(X, \Omega_X^1)$.

In 1934, Chevalley and Weil formulated in [1] a formula for the case that $k = \mathbb{C}$, which has been proved by Weil in [9]. In this thesis the same formula will be proved for any algebraically closed field k with $\text{char } k \nmid \#G$. In the 80s of the 20th century, Nakajima [6, 7] and Kani [4] gave a far more general result.

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Introduction

Let $f: X \rightarrow Y$ be a Galois covering of complete, nonsingular algebraic curves over an algebraically closed field k . Its Galois group $G = \text{Cov}(X/Y)$ acts in a natural way on the space of regular 1-forms on X , thus we are given a linear representation $G \rightarrow \text{GL}(H^0(X, \Omega_X^1))$. The main question in this thesis is how many times a given irreducible representation of G occurs in $H^0(X, \Omega_X^1)$.

In the 30s of the 20th century, Chevalley and Weil solved this problem for the case of Riemann surfaces, thus for $k = \mathbb{C}$. They stated their formula in [1], but gave a proof only for the case that f is unramified. For this *Speziellfall*, they found that the space consists of one trivial representation and $g_Y - 1$ times the regular representation, thus

$$H^0(X, \Omega_X^1) \cong k \oplus k[G]^{\oplus(g_Y-1)}$$

as $k[G]$ -modules, where k denotes the trivial $k[G]$ -module. The article was followed by [9], where the formula was proven for general holomorphic f .

Their proof consists of two parts. First, a generalization of the Riemann-Roch Theorem is proved, which is used in the second part to prove the main theorem. In the second part, an *unendlichvielblättrigen Überlagerungsfläche* $\mathfrak{U}$ of Y is introduced, where the calculations are done with a corresponding *Grenzkreisuniformisierende* ω .

In 1984, Nakajima [6] considered this problem on a more general level. For projective varieties X and Y of arbitrary dimension, a finite *étale* Galois covering $f: X \rightarrow Y$ with group G , and any coherent sheaf $\mathcal{G}$ on Y , he investigated the $k[G]$ -module structure of the cohomology groups $H^i(X, f^*\mathcal{G})$. In the case of curves, which he discussed explicitly in §3, we may choose the sheaf to be Ω_Y^1 , and since f is *étale*, hence unramified, we have $f^*\Omega_Y^1 = \Omega_X^1$. Indeed, as a corollary he found the above equation for the case that $\text{char } k \nmid \#G$.

Two years later, Nakajima [7] generalized his results for any tamely ramified Galois covering f of projective varieties. In the same year, Kani [4] published an article that has some overlap of results, unfortunately he was unaware of the work by Nakajima. The paper of Kani only deals with a special case of Nakajima's results, it concerns tamely ramified Galois coverings $f: X \rightarrow Y$ of curves and its aim is to completely determine the structure of $H^0(X, \Omega_X^1)$ as a $k[G]$ -module, for any characteristic of the field k .

This thesis discusses a case that is slightly more general than that of Chevalley and Weil, namely for k any algebraically closed field with $\text{char } k \nmid \#G$. Kani [4] writes that “it is immediate” that their classical result remains valid for such a field k , so roughly speaking this thesis comes down to repeating the proof of 1934, using 1970s-techniques (e.g., Hartshorne [3] and Fulton [2]). The proof given here, actually resembles a lot to the classical one. Instead of proving a generalization of Riemann-Roch, we simply use an existing one from Fulton, which is Hirzebruch-Riemann-Roch (Thm. 1.2) and instead of the huge covering $\mathcal{U} \rightarrow Y$, we elegantly use the fibred product $X \times_Y X \rightarrow Y$ (see Section 2.4).

In Chapter 1 we will discuss some general concepts that we will use in the proof, like the Hirzebruch-Riemann-Roch Theorem mentioned above and the theory of separable extensions of function fields in §1.1, the Frobenius Reciprocity formula in §1.2, the action of the Galois group on the sheaf $f_*\mathcal{O}_X$ in §1.3 and the concept of local monodromy in §1.4. Be aware of the fact that these sections are meant to be preparatory.

In Chapter 2 we will prove the Chevalley-Weil formula for the case $\text{char } k \nmid \#G$. As a side-effect, we will prove the Riemann-Hurwitz formula in §2.2, come to a preliminary version of the main formula in §2.3 and finally finish the proof using the fibred product $X \times_Y X \rightarrow Y$ in §2.4. After completing the proof, we may take a breath by looking at some examples in Section 2.5.

Who should read this thesis? Imagine a graduate student from any city on the globe, who is more or less familiar with the algebraic geometry of nonsingular curves, and just found this document on the Internet by looking for some information about the Chevalley-Weil formula. Well, this student is the lucky one. Please contact me if you are that student, I would be very happy if you actually exist.

I expect the reader to be familiar with basic algebraic geometry, as described in Hartshorne [3], Chapter I, II and IV, and some elementary linear representation theory, as described in Serre [5], Part I. Of course other areas of mathematics, like Galois theory for instance, will be used, but it is not a catastrophe if the reader is not familiar with it.

Perhaps a more important question is *how* one should read this thesis. I would recommend —certainly if you are not the imaginary student described above— to start with the *Samenvatting voor de eerstejaars student*, followed by the examples in Section 2.5, and then read the Introduction again, so that it is really clear what the main question is about. After that, you are free to read whatever you want.

Chapter 1

Preliminaries

This chapter consists of preparatory material we will need in order to prove the Chevalley-Weil formula. Do not expect this chapter to be a comprehensive introduction to the different areas of mathematics. I would rather recommend the reader to go as quickly as possible to Chapter 2, and go back if some theory is not clear.

1.1 Algebraic Geometry

In this thesis we will use the same definitions and conventions as used in Hartshorne [3], IV §1. This means that we will use the word *curve* for a complete, nonsingular curve over an algebraically closed field k . To be precise, a curve is an integral scheme of dimension 1, proper over k , all of whose local rings are regular. We denote the genus of a curve X by g_X .

A morphism $f: X \rightarrow Y$ is always finite and separable, with only tame ramification. We demand that $\text{char } k \nmid \deg(f)$.

Since the curve X is nonsingular, we have by Hartshorne [3], Prop. II 6.10 an isomorphism between the group $\text{Cl}(X)$ of (Weil) divisors modulo linear equivalence and the group $\text{Pic}(X)$ of invertible sheaves modulo isomorphism (c.f. Hartshorne [3], IV §1). This motivates the following definitions.

Definition We define the *degree* of an invertible sheaf $\mathcal{L}$ on the curve X as the degree of its corresponding divisor. If $\mathcal{E}$ is a locally free $\mathcal{O}_X$ -module of rank r , then we define the *determinant* as the r -th exterior power of $\mathcal{E}$, thus $\det \mathcal{E} := \bigwedge^r \mathcal{E}$, which is an invertible sheaf. We define the degree of $\mathcal{E}$ to be the degree of $\det \mathcal{E}$. We also write $\int_X c_1(\mathcal{E})$ instead of $\deg \mathcal{E}$.

Let us state some properties of this degree we will use in a later stage. If $\mathcal{L}$ is an invertible sheaf on X and $\mathcal{E}$ is a locally free $\mathcal{O}_X$ -module of rank r , then we have the following relation,

$$\deg(\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{E}) = (\text{rank } \mathcal{E}) \deg(\mathcal{L}) + \deg(\mathcal{E}),$$

since we have that $\det(\mathcal{L} \otimes \mathcal{E}) = \bigwedge^r(\mathcal{L} \otimes \mathcal{E}) = \mathcal{L}^{\otimes r} \otimes \det \mathcal{E}$.

Another property is stated in next lemma.

Lemma 1.1 *Let $f: X \rightarrow Y$ be a finite morphism of curves and $\mathcal{G}$ a locally free $\mathcal{O}_Y$ -module, then we have*

$$\deg(f^*\mathcal{G}) = \deg(f) \deg(\mathcal{G}).$$

Proof. By the isomorphism mentioned above ([3] Prop. II 6.10), we may apply Hartshorne [3], Prop. II 6.9 for the case that $\mathcal{G}$ is an invertible sheaf. The general case follows because $f^* \det \mathcal{G} = \det f^* \mathcal{G}$. $\square$

Definition Let X be a curve and $\mathcal{F}$ be a coherent sheaf on X , then we define the *Euler Characteristic* of $\mathcal{F}$ by

$$\text{EC}(X, \mathcal{F}) := \dim_k H^0(X, \mathcal{F}) - \dim_k H^1(X, \mathcal{F}).$$

Remark The Euler Characteristic is usually defined as an infinite alternating sum over all its cohomology groups. However, for curves this is the same as the definition above, due to Grothendieck's Vanishing Theorem (Hartshorne [3], Thm. III 2.7).

Theorem 1.2 (Riemann-Roch) *Let $\mathcal{F}$ be a locally free $\mathcal{O}_X$ -module. Then the Euler Characteristic is equal to*

$$\text{EC}(X, \mathcal{F}) = (\text{rank } \mathcal{F})(1 - g_X) + \int_X c_1(\mathcal{F}).$$

Proof. See Fulton [2], Cor. 15.2.1 (Hirzebruch-Riemann-Roch) and Example 15.2.1. $\square$

Let $\mathcal{F}$ be again a sheaf on a curve X . We define the *support* of $\mathcal{F}$ to be the set of all points of X where the stalk $\mathcal{F}_P$ is not equal zero. In a later stage, we will deal with sheaves that have a support that only consists of finitely many closed points. The following lemma gives a property of its cohomology groups.

Lemma 1.3 *Let $\mathcal{F}$ be a sheaf on a curve X with a support that is a finite set of closed points. Then we have*

$$H^0(X, \mathcal{F}) = \bigoplus_{P \in \text{Supp}(\mathcal{F})} \mathcal{F}_P \quad \text{and} \quad H^1(X, \mathcal{F}) = 0.$$

Proof. Let $i: \text{Supp}(\mathcal{F}) \hookrightarrow X$ be the inclusion map. By adjunction of i^{-1} and i^* , we have a natural map $\mathcal{F} \rightarrow i_* i^{-1} \mathcal{F}$. We prove that this map is an isomorphism. It suffices to show that it is an isomorphism on stalks. If $P \notin \text{Supp}(\mathcal{F})$, we have $\mathcal{F}_P = 0$ and $(i_* i^{-1} \mathcal{F})_P = 0$. If $P \in \text{Supp}(\mathcal{F})$, then

$$(i_* i^{-1} \mathcal{F})_P = \lim_{U \ni P} (i^{-1} \mathcal{F})(U \cap \text{Supp}(\mathcal{F})) = (i^{-1} \mathcal{F})_P = \mathcal{F}_P,$$

where the limit is taken over the open sets U containing P ; this is because the sets of the form $U \cap \text{Supp}(\mathcal{F})$ are precisely the open neighbourhoods of P in $\text{Supp}(\mathcal{F})$.

The assertion that $H^1(X, \mathcal{F}) = 0$ is a special case of Grothendieck's Vanishing Theorem, see Hartshorne [3], Thm. III 2.7 and Lemma 2.10. $\square$

At this point, two lemmas about the functor $\mathcal{F} \mapsto f_*\mathcal{F}$ are included.

Lemma 1.4 *Let $f: X \rightarrow Y$ be a finite separable morphism of curves. Then the functor*

$$\mathcal{F} \mapsto f_*\mathcal{F}$$

on the category of coherent sheaves on X is exact.

Proof. If $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ is a short exact sequence of coherent sheaves on X then we get a long exact sequence

$$0 \rightarrow f_*\mathcal{F} \rightarrow f_*\mathcal{G} \rightarrow f_*\mathcal{H} \rightarrow R^1f_*\mathcal{F}$$

on Y . But f is finite, hence affine, so $R^1f_*\mathcal{F} = 0$ as readily follows from Hartshorne [3], III Thm. 3.7 and Prop. 8.1. $\square$

Lemma 1.5 *Let $f: X \rightarrow Y$ be a finite morphism of curves and let $\mathcal{F}$ be a coherent sheaf on X . Then there exists a natural isomorphism of cohomology groups $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$.*

Proof. Since any finite morphism is affine (Hartshorne [3], Ex. II 5.17b) and our curves X and Y are noetherian and separated, the lemma follows from Hartshorne [3], Ex. III 4.1. $\square$

Separable Extensions of Function Fields

Let us very quickly recall some facts about nonsingular curves and separable field extensions. The reason why we do this, except for its beauty, is that it will be used in the proofs of Lemmas 1.6 and 1.13.

Let Q be a point on a nonsingular curve Y . According to Hartshorne [3], Thms. I 5.1 & 6.2A, its local ring $\mathcal{O}_{Y,Q}$ is a *Dedekind domain*, i.e., an integrally closed noetherian domain of dimension one. Because it is local too, it is also a discrete valuation ring. Its valuation v_Q sends an element $y \in \mathcal{O}_{Y,Q}$ to the integer l for which we have that $y \in \mathfrak{m}_Q^l - \mathfrak{m}_Q^{l+1}$, where $\mathfrak{m}_Q$ denote the maximal ideal of $\mathcal{O}_{Y,Q}$. The quotient field of $\mathcal{O}_{Y,Q}$ is equal to the function field $K(Y)$.

Due to the morphism f , we are given a finite separable field extension $K(X)/K(Y)$, of degree $n = \deg(f)$. Let B denote the integral closure of $\mathcal{O}_{Y,Q}$ in $K(X)$. By Hartshorne [3], Thm. I 6.3, this is again a Dedekind domain. This ring is not local in general. Its maximal ideals precisely correspond to the points $P_1, \dots, P_m$ on X that lie above Q . If we write $\mathfrak{m}_{P_j}$ for the ideal corresponding to the point P_j , then we have $\mathfrak{m}_Q = \mathfrak{m}_{P_j} \cap \mathcal{O}_{Y,Q}$ for every $j = 1, \dots, m$. In $K(X)$, the ideal $\mathfrak{m}_Q$ (by which we mean the ideal $\mathfrak{m}_QB$) is not maximal anymore and because B is Dedekind, it can be written uniquely as $\mathfrak{m}_Q = \prod_{j=1}^m \mathfrak{m}_{P_j}^{e_j}$, where e_j is called the *ramification index* of P_j . See Figure 1.1.

$$\begin{array}{ccccc}
K(X) & \supseteq & B & \supseteq & \prod_{j=1}^m \mathfrak{m}_{P_j}^{e_j} \\
| & & | & & | \\
K(Y) & \supseteq & \mathcal{O}_{Y,Q} & \supseteq & \mathfrak{m}_Q
\end{array}$$

Figure 1.1: The field extension $K(X)/K(Y)$

Lemma 1.6 *Let $f: X \rightarrow Y$ be a finite separable morphism of curves, and let Q be an arbitrary closed point on Y . Then we have*

$$\deg(f) = \sum_{P \in f^{-1}(Q)} e_P,$$

where e_P denotes the ramification index of f at P .

Proof. Let n denote the degree of f and write $\{P_1, \dots, P_m\}$ for the points lying above Q . We have by Serre [8], Prop. I 10, the relation $n = \sum_j e_j f_j$, where f_j is called the *inertia index* of f at P_j . It is by definition equal to the degree of the extension of the residue fields,

$$f_j := [B/\mathfrak{m}_{P_j} : \mathcal{O}_{Y,Q}/\mathfrak{m}_Q] = [k(P_j) : k(Q)].$$

In our case, this is always one, since our curves are nonsingular over an algebraically closed field and thus $k(P) = k$ for any P . $\square$

Let us fix some $P = P_j$. The localization of B at the maximal ideal $\mathfrak{m}_P$ gives the local ring of the corresponding point on X , $B_{\mathfrak{m}_P} = \mathcal{O}_{X,P}$. This is again a discrete valuation ring, with valuation v_P , such that we obtain an extension of discrete valuation rings, which is given as follows. Let t be a *uniformizer* or *local parameter* of $\mathcal{O}_{Y,Q}$, i.e., an element with $v_Q(t) = 1$, and let z be a uniformizer of $\mathcal{O}_{X,P}$. Then the extension is given by an injective homomorphism

$$\mathcal{O}_{Y,Q} \rightarrow \mathcal{O}_{X,P}, \quad t = uz^{e_P},$$

where u is some unit of $\mathcal{O}_{X,P}$. We say that the valuation v_P *prolongs* v_Q with index e_P .

1.2 Linear Representation Theory

Remember that k is an algebraically closed field. Let G be a finite group of order n with $\text{char } k \nmid n$ and let the set $\{\chi_1, \dots, \chi_h\}$ be the set of the irreducible characters of G . We denote by d_χ the dimension over k of an irreducible G -representation with character χ .

Let us be given a certain linear G -representation $G \rightarrow \text{GL}(V)$. Under this map, we consider the k -vector space V as a $k[G]$ -module. For an irreducible character χ , we define an element

$$p_\chi := \frac{d_\chi}{n} \sum_{\sigma \in G} \chi(\sigma^{-1}) \sigma$$

of the group algebra $k[G]$. According to Serre [5], §6.3 Ex. 6.4, we find that they satisfy the relations $p_\chi^2 = p_\chi$ and $p_\chi p_{\chi'} = 0$ for $\chi \neq \chi'$. By [5] §2.6, Thm. 8, such an element acts as a k -linear map on V and its image is precisely the χ -isotypic component of V , which we denote by V^χ . Thus we have a decomposition of V as a direct sum of its isotypic components, $V = \bigoplus_\chi V^\chi$.

Given an irreducible representation $\rho_\chi: G \rightarrow \mathrm{GL}(W)$, then we denote its *multiplicity* in some G -representation V by $\langle \chi, V \rangle_G$. Thus we define

$$\langle \chi, V \rangle_G := \dim_k \mathrm{Hom}_{k[G]}(W, V).$$

Note that the dimension of an isotypic component V^χ is equal to the dimension of χ , i.e., the dimension of W , times the multiplicity of χ . In a formula, $\dim_k V^\chi = d_\chi \langle \chi, V \rangle_G$.

We will often use the following formula. We assume that the reader is familiar with the notion of an *induced representation*.

Proposition 1.7 (Frobenius Reciprocity Formula) *Given a finite group G with $\mathrm{char} k \nmid \#G$ and let H be a subgroup of G . Let V be a G -representation that is induced by the H -representation W , thus $V = \mathrm{Ind}_H^G W$. Then the multiplicity of an irreducible character χ of G in V is given by*

$$\langle \chi, V \rangle_G = \langle \mathrm{Res}_H^G \chi, W \rangle_H.$$

Proof. Serre [5] §7.2, Thm. 13. □

Recall that a *fundamental character* η of a cyclic group G_P is a character that has the property that $\eta(G_P) = \{\zeta_e^\alpha\}$, where ζ_e^α denotes a power of a primitive e -th root of unity. Equivalently, η is a fundamental character if any character ξ of G_P can be written as a power of η .

1.3 Galois coverings

Let us briefly recall some facts from Galois Theory. Let $f: X \rightarrow Y$ be a finite separable morphism of curves. We call f a *Galois covering* if the field extension of its function fields $K(X)/K(Y)$ is Galois. This is equivalent with the property that the group of covering transformations

$$\mathrm{Cov}(X/Y) := \{\sigma: X \rightarrow X \mid f \circ \sigma = f\}$$

acts transitively on the fibre of f at every point Q on Y . In other words, if $P_1, P_2 \in f^{-1}(Q)$, then there is an element $\sigma \in G$, such that $P_1 = \sigma P_2$. If the covering X/Y is Galois, we call $\mathrm{Cov}(X/Y)$ the *Galois group* of the covering. It is a fact from Galois Theory that $\#G = \deg(f)$ if and only if X/Y is Galois.

From now on, we assume that $f: X \rightarrow Y$ is a Galois covering of degree n and with Galois group G . We furthermore assume that $\mathrm{char} k \nmid \#G$.

Let $\mathcal{F}$ be a sheaf on X , then the Galois group G acts in a natural way on the sheaf $f_*\mathcal{F}$. Let σ be an element of G and let ϕ be an element of $f_*\mathcal{F}$ (defined

on an open subset $V \subseteq Y$ or as an element of the stalk at some point on Y), then we define the action of σ on ϕ as a left action

$$\sigma * \phi := (\sigma^{-1})^* \phi = \phi \circ \sigma^{-1}.$$

This is well-defined because σ is an automorphism of $f^{-1}(V)$.

Since the morphism f is surjective, we know that we have the following exact sequence of sheaves on Y ,

$$0 \rightarrow \mathcal{O}_Y \xrightarrow{f^\#} \mathcal{O}_X.$$

If P is a point on X and $Q = f(P)$ is a point on Y , then for any local parameter t in $\mathcal{O}_{Y,Q}$ and any local parameter z in $\mathcal{O}_{X,P}$, we have an inclusion

$$f_P^\# : \mathcal{O}_{Y,Q} \hookrightarrow \mathcal{O}_{X,P}, \quad t = uz^{e_P},$$

where $u \in \mathcal{O}_{X,P}^*$ is a unit and e_P is the ramification index of f at P . Under this homomorphism, we can consider $\mathcal{O}_{X,P}$ as an $\mathcal{O}_{Y,Q}$ -module.

Let us consider the completion of the local ring $\mathcal{O}_{X,P}$. Because X and Y are nonsingular curves, we may take as identifications $\hat{\mathcal{O}}_{Y,Q} \cong k[[t]]$ and $\hat{\mathcal{O}}_{X,P} \cong k[[z]]$. The canonical homomorphism can be extended to these complete local rings, which are easier to calculate with.

Lemma 1.8 *Let $f: X \rightarrow Y$ be a finite morphism, let $P \in X$ and set $Q = f(P)$. Let e_P be the ramification index of f at P and assume that e_P is prime to char k . Let t be a local parameter in $\mathcal{O}_{Y,Q}$. Then there exists a local parameter $z \in \hat{\mathcal{O}}_{X,P}$ such that the canonical homomorphism $\hat{\mathcal{O}}_{Y,Q} \rightarrow \hat{\mathcal{O}}_{X,P}$ is given by $t \mapsto z^{e_P}$.*

Proof. Choose a uniformizer $w \in \mathcal{O}_{X,P}$. Let the canonical homomorphism $\hat{\mathcal{O}}_{Y,Q} = k[[t]] \rightarrow \hat{\mathcal{O}}_{X,P} = k[[w]]$ be given by $t \mapsto uw^e$ with $u \in k[[w]]^*$ and $e = e_P$. If we can find a unit $v \in k[[w]]^*$ with $v^e = u$ then we are done, because $z := vw$ is again a local parameter at P and $z^e = t$.

Because k is algebraically closed we may assume that the constant term of u is equal to 1. Writing $\mathfrak{m} = (w)$ for the maximal ideal of $k[[w]]$, it suffices to show that the map $a \mapsto a^e$ is surjective on $1 + \mathfrak{m}$.

We are going to construct a sequence (a_n) in $1 + \mathfrak{m}$ such that a_n^e converges to u in the $\mathfrak{m}$ -adic topology. As $1 + \mathfrak{m} \subset k[[w]]^*$ is complete for the $\mathfrak{m}$ -adic topology, this suffices. We construct the sequence (a_n) inductively in such a way that a_n^e is congruent to u modulo $\mathfrak{m}^n$. We start with $a_1 = 1$. Now suppose that $n \geq 2$ and that we already have an element $a_n \in 1 + \mathfrak{m}$ such that $v_P(u - a_n^e) \geq n$, which just means that we can write

$$u - a_n^e = w^n \cdot (c_0 + c_1w + c_2w^2 + \cdots)$$

with coefficients $c_i \in k$. Because e is assumed to be prime to char k we can define $d_0 = c_0/e$. Let $a_{n+1} = a_n + d_0w^n$. Then

$$a_{n+1}^e \equiv a_n^e + a_n^{e-1} \cdot c_0w^n \pmod{w^{n+1}}$$

and because $a_n \equiv 1 \pmod{w}$ we find that $a_{n+1}^e \equiv a_n^e + c_0 w^n \pmod{w^{n+1}}$. So $v_P(u - a_{n+1}^e) \geq n + 1$, which is what we wanted. $\square$

Lemma 1.9 *Let $f: X \rightarrow Y$ be a Galois covering and let $P \in X$ and $Q = f(P)$. Then the automorphism group of $\hat{\mathcal{O}}_{X,P}$ as an algebra over $\hat{\mathcal{O}}_{Y,Q}$ is cyclic of order e_P .*

Proof. According to Lemma 1.8 we have, after a suitable choice of local coordinates, that the homomorphism $\hat{\mathcal{O}}_{Y,Q} = k[[t]] \rightarrow \hat{\mathcal{O}}_{X,P} = k[[z]]$ is given by $t = z^e$. Then it follows immediately that the automorphisms of $\hat{\mathcal{O}}_{X,P}$ as an $\hat{\mathcal{O}}_{Y,Q}$ -algebra are the automorphisms given by $z \mapsto \zeta \cdot z$ where ζ is an e -th root of unity. Hence the automorphism group is cyclic of order e . $\square$

Definition The stabilizer group G_P of a point $P \in X$ is defined as

$$G_P = \{\sigma \in G \mid \sigma(P) = P\}.$$

The following lemma is a basic fact from Galois Theory and elementary Group Theory.

Lemma 1.10 *Let $f: X \rightarrow Y$ be a Galois covering and let Q be a point on Y . Then all stabilizer groups G_{P_j} with $P_j \in f^{-1}(Q)$ have the same order. In particular, we have the equality $\#G_P = \#G/\#f^{-1}(Q)$.*

Proof. Since G is a finite group, we may denote the index of the subgroup G_P in G by $[G : G_P] = \#G/\#G_P$. If we denote the orbit of the point P under G by GP , then by Group Theory we have $\#G = (\#G_P)(\#GP)$. Since X/Y is Galois, the group G acts transitively on the fibre, so we get $GP = f^{-1}(Q)$, which is independent of the chosen point $P \in f^{-1}(Q)$. This proves the lemma. $\square$

Proposition 1.11 *Let $f: X \rightarrow Y$ be a Galois covering. The stabilizer group G_P of a point P on X is cyclic of order e_P .*

Proof. The stabilizer group G_P has a natural action on $\mathcal{O}_{X,P}$, so it acts on $\hat{\mathcal{O}}_{X,P}$ by $\hat{\mathcal{O}}_{Y,Q}$ -automorphisms. It follows that we have an embedding

$$G_P \hookrightarrow \text{Aut}_{\hat{\mathcal{O}}_{Y,Q}}(\hat{\mathcal{O}}_{X,P}),$$

so G_P is a cyclic group of order dividing e_P . In particular we have $\#G_P \leq e_P$.

Now we calculate the degree of f by Lemma 1.6 and due to Lemma 1.10, we find

$$\deg(f) = \sum_{P \in f^{-1}(Q)} e_P \geq \sum_{P \in f^{-1}(Q)} \#G_P = [G : G_{P_0}] \#G_{P_0} = \#G.$$

Therefore this inequality is in fact an equality, thus we find $\#G_P = e_P$. $\square$

Remark From Lemma 1.10 and Proposition 1.11, we may draw the conclusion that all points in some fibre $f^{-1}(Q)$ have the same ramification index e_P , with $e_P = \deg(f)/\#f^{-1}(Q)$ for any P in $f^{-1}(Q)$, and that their stabilizer groups are conjugate via $G_{\sigma P} = \sigma^{-1}G_P\sigma$.

Proposition 1.12 *Let $f: X \rightarrow Y$ be a Galois covering, let P be a point on X and set $Q = f(P)$. Then $\mathcal{O}_{X,P}$ is a free of rank e_P as a module over $\mathcal{O}_{Y,Q}$.*

Proof. The ring $\mathcal{O}_{Y,Q}$ is a discrete valuation ring, and $\mathcal{O}_{X,P}$ is finitely generated as a module over $\mathcal{O}_{Y,Q}$ because the morphism f is finite. Moreover it is clear that $\mathcal{O}_{X,P}$, as an $\mathcal{O}_{Y,Q}$ -module, is torsion-free. It then follows from the general theory of modules over a principal ideal domain that $\mathcal{O}_{X,P}$ is free over $\mathcal{O}_{Y,Q}$, say of rank r . To see that $r = e_P$ we may pass to $\mathfrak{m}_Q$ -adic completions. But the $\mathfrak{m}_Q$ -adic completion of $\mathcal{O}_{X,P}$ is just $\hat{\mathcal{O}}_{X,P}$, because $\mathfrak{m}_Q \cdot \mathcal{O}_{X,P} = (\mathfrak{m}_P)^e$ is an ideal of definition for the $\mathfrak{m}_P$ -adic topology. Now it is clear from Prop. 1.8 that $\hat{\mathcal{O}}_{X,P}$ is free of rank e as a module over $\hat{\mathcal{O}}_{Y,Q}$. $\square$

Lemma 1.13 *Let P be a point on X , set $Q = f(P)$ and let G_P be the stabilizer group of P . Then we have the equality*

$$\mathcal{O}_{X,P}^{G_P} = \mathcal{O}_{Y,Q}.$$

Proof. The proof is in fact a continuation of the theory about separable field extensions and an extension of Figure 1.1.

We are given a field extension $K(X)/K(Y)$ that is Galois with group G . Note that the ideal $\mathfrak{m}_P$ is stable under the stabilizer group $G_P \subseteq G$. (In [8] §7, Serre calls G_P the *decomposition group* of $\mathfrak{m}_P$ in $K(X)/K(Y)$.) We define $K(X)^{G_P}$ to be the intermediate field of all elements of $K(X)$ that are invariant under G_P . It is a field extension of $K(Y)$, which is Galois if and only if G_P is a normal subgroup of G .

Let again B be the integral closure of $\mathcal{O}_{Y,Q}$ in $K(X)$ and let us write C for the integral closure of $\mathcal{O}_{Y,Q}$ in $K(X)^{G_P}$. The ring B contains the maximal ideal $\mathfrak{m}_P$ and we have $\mathfrak{m}_Q = \mathfrak{m}_P \cap \mathcal{O}_{Y,Q}$. We define $\mathfrak{n} = \mathfrak{m}_P \cap C$, which is a maximal ideal in C . By construction, we have $B_{\mathfrak{m}_P} = \mathcal{O}_{X,P}$ and $\mathcal{O}_{X,P}^{G_P} = C_{\mathfrak{n}}$. It follows that we have extensions of discrete valuation rings,

$$\begin{array}{ccccc} \mathcal{O}_{Y,Q} & \rightarrow & \mathcal{O}_{X,P}^{G_P} & \rightarrow & \mathcal{O}_{X,P} \\ t & \mapsto & u's^h & \mapsto & uz_P^e, \end{array}$$

where t , s and z are uniformizers and u and u' are units of the corresponding local ring. We see that v_P prolongs v_Q with index e_P and that $v_{\mathfrak{n}}$ prolongs v_Q with index h . Since all residue fields are equal to k itself, the residue extension is trivial and it follows by Serre [8], Prop. I 21b, that the index h is equal to one. Thus $\mathcal{O}_{X,P}^{G_P} = \mathcal{O}_{Y,Q}$. $\square$

Recall the notion of a fundamental character, c.f. §1.2.

Lemma 1.14 *Let $f: X \rightarrow Y$ be a Galois covering and let P be a point on X . The Zariski cotangent space $\mathfrak{m}_P/\mathfrak{m}_P^2$ is an irreducible G_P -representation with a fundamental character.*

Proof. Write $\mathfrak{m}_P \subseteq \mathcal{O}_{X,P}$ for the maximal ideal of P . The ideals $\mathfrak{m}_P^i$ can be seen as infinite dimensional k -vector spaces. Note that the natural map $k \rightarrow k(P) = \mathcal{O}_{X,P}/\mathfrak{m}_P$ is an isomorphism, since k is algebraically closed. The group G_P acts on these spaces and because $\text{char } k \nmid \#G_P$, the spaces are semisimple:

$$\mathfrak{m}_P^i = \bigoplus_{\xi} [\mathfrak{m}_P^i]^{\xi},$$

where ξ runs over all irreducible representations of G_P .

Consider the Zariski cotangent space $\mathfrak{m}_P/\mathfrak{m}_P^2$, which is a one dimensional k -vector space, where we again consider k as the residue field $\mathcal{O}_{X,P}/\mathfrak{m}_P$. Since the ideals $\mathfrak{m}_P^i$ are stable under G_P , the quotient $\mathfrak{m}_P/\mathfrak{m}_P^2$ is G_P -stable as well. Since it has dimension one, it must be an irreducible representation of G_P , say with character η . This means that for this character η , we have a strict inclusion $[\mathfrak{m}_P^2]^{\eta} \subsetneq [\mathfrak{m}_P]^{\eta}$, while for all other irreducible representations ξ , we have an equality $[\mathfrak{m}_P^2]^{\xi} = [\mathfrak{m}_P]^{\xi}$. Therefore we can pick a local parameter z in $\mathfrak{m}_P - \mathfrak{m}_P^2$, which is in the η -eigenspace. This means that the group G_P acts on this z as $\tau * z = (\tau^{-1})^* z = \eta(\tau)z$ for $\tau \in G_P$.

Now we must show that this η is a fundamental character. By looking at valuations it is clear that the elements $\{1, z, \dots, z^{e_P-1}\}$ are linearly independent over $\mathcal{O}_{Y,Q}$. If η is not a fundamental character, then we have a strict divisor $d \mid e$ with $\eta^d(\tau) = 1$ for all $\tau \in G_P$. But then 1 and z^d are both G_P -invariant, and this is in contradiction with Lemma 1.13. $\square$

Proposition 1.15 *Let $f: X \rightarrow Y$ be a Galois covering, let P be a point on X and set $Q = f(P)$. Denote the fundamental character of the Zariski cotangent space by η . Then there exist local parameters $t \in \mathcal{O}_{Y,Q}$ and $z \in [\mathcal{O}_{X,P}]^{\eta}$, such that we can make the following claims.*

- (i) *The canonical inclusion $f^{\#}: \mathcal{O}_{Y,Q} \hookrightarrow \mathcal{O}_{X,P}$ is given by $t \mapsto z^{e_P}$.*
- (ii) *A basis for $\mathcal{O}_{X,P}$ as a free $\mathcal{O}_{Y,Q}$ -module is given by*

$$\{1, z, z^2, \dots, z^{e_P-1}\}.$$

We may consider this as a decomposition of $\mathcal{O}_{X,P}$ into its isotypic components, where for any irreducible character η^{α} of G_P , the η^{α} -eigenspace is given by

$$[\mathcal{O}_{X,P}]^{\eta^{\alpha}} = \mathcal{O}_{Y,Q} \cdot z^{\alpha}.$$

Proof. (i) By Lemma 1.14 there exists a fundamental character η of G_P and a local parameter $z \in (\mathcal{O}_{X,P})^{\eta}$. Let t be a local parameter in $\mathcal{O}_{Y,Q}$, then $t = uz^e$ via the canonical homomorphism, where $e = e_P$. Let τ be an element of G_P , then we have $\tau * t = t$ and $\tau * (uz^e) = (\tau^{-1})^*(u)\eta^e(\tau)z^e = (\tau^{-1})^*(u)z^e$. It follows that the unit u is invariant under G_P , thus by Lemma 1.13 it is an element of $\mathcal{O}_{Y,Q}$. Now replace t by $u^{-1}t$ and the result follows.

(ii) Because $t \mapsto z^{e_P}$ it is clear that $\{1, z, z^2, \dots, z^{e_P-1}\}$ is a basis of $\mathcal{O}_{X,P}$ as a module over $\mathcal{O}_{Y,Q}$. Moreover, it is clear that G_P acts on $\mathcal{O}_{Y,Q} \cdot z^i$ through the character η^i . As η is a fundamental character it follows that $\mathcal{O}_{X,P} = \bigoplus_{i=0}^{e_P-1} \mathcal{O}_{Y,Q} \cdot z^i$ is the decomposition of $\mathcal{O}_{X,P}$ into isotypic components. $\square$

Notice that every irreducible representation of G_P occurs precisely once in $\mathcal{O}_{X,P}$ as $\mathcal{O}_{Y,Q}$ -module.

Proposition 1.16 *Let $f: X \rightarrow Y$ be a Galois covering of degree n . We can make the following statements about the sheaf $f_*\mathcal{O}_X$.*

- (i) *The sheaf $f_*\mathcal{O}_X$ is a locally free $\mathcal{O}_Y$ -module of rank n .*
- (ii) *For an irreducible character χ of the group G with degree d_χ , the isotypic component $(f_*\mathcal{O}_X)^\chi$ is a locally free $\mathcal{O}_Y$ -module of rank d_χ^2 .*
- (iii) *The inclusion map $j: \mathcal{O}_Y \hookrightarrow (f_*\mathcal{O}_X)^G$ is an isomorphism.*

Proof. (i) If $U \subset Y$ is any affine open subset then $A := \mathcal{O}_Y(U)$ is a Dedekind ring and $B := \mathcal{O}_X(f^{-1}(U))$ is finitely generated and torsion-free as an A -module. By the general theory of modules over a Dedekind ring, it follows that B is projective (i.e., locally free) as an A -module. Hence $f_*\mathcal{O}_X$ is a locally free $\mathcal{O}_Y$ -module. Its rank is by definition the degree of f .

(ii) We have a direct sum decomposition $f_*\mathcal{O}_X = \bigoplus_\chi (f_*\mathcal{O}_X)^\chi$, so each $f_*\mathcal{O}_X^\chi$ is again locally free, say of rank r_χ . If $Q \in Y$ then we have $(f_*\mathcal{O}_X)_Q = \bigoplus_\chi ((f_*\mathcal{O}_X)_Q)^\chi$. Now we pass to $\mathfrak{m}_Q$ -adic completions. The $\mathfrak{m}_Q$ -adic completion of $(f_*\mathcal{O}_X)_Q$ is $\bigoplus_{P \in f^{-1}(Q)} \hat{\mathcal{O}}_{X,P}$, so we find that the completed stalk of $(f_*\mathcal{O}_X)^\chi$ at Q is $(\bigoplus_{P \in f^{-1}(Q)} \hat{\mathcal{O}}_{X,P})^\chi$, and we know that this is a free $\hat{\mathcal{O}}_{Y,Q}$ -module of rank r_χ . Now take for Q any point which is not a branch point for the morphism f , and fix a point $P \in f^{-1}(Q)$. Then $f^{-1}(Q)$ is in bijection with G under the map $g \mapsto gP$. For each $g \in G$ the natural map $\hat{\mathcal{O}}_{Y,Q} \rightarrow \hat{\mathcal{O}}_{X,gP}$ is an isomorphism, and an element $h \in G$ acts on $(f_*\mathcal{O}_X)_Q^\chi = \bigoplus_{g \in G} \hat{\mathcal{O}}_{X,gP}^\chi$ via the collection of isomorphisms $(h^{-1})^*: \hat{\mathcal{O}}_{X,gP}^\chi \xrightarrow{\sim} \hat{\mathcal{O}}_{X,hgP}^\chi$. This just says that $(f_*\mathcal{O}_X)_Q^\chi \cong \text{Reg}_G \otimes_k \hat{\mathcal{O}}_{Y,Q}^\chi$, where Reg_G is the regular representation of G . It follows that $r_\chi = d_\chi^2$.

(iii) Take an affine open $U \subset Y$, and write $A = \mathcal{O}_Y(U)$ and $B = \mathcal{O}_X(f^{-1}(U))$. Then B is a subring of the function field $k(X)$, so $B^G \subset k(X)^G = k(Y)$. Further, B is integral over A and A is integrally closed in $k(Y)$, so the inclusion $A \subset B^G$ is an isomorphism. $\square$

Remark If we denote the order of G by n , then we have indeed

$$\text{rank } f_*\mathcal{O}_X = \text{rank } \bigoplus_\chi f_*\mathcal{O}_X^\chi = \sum_\chi \text{rank } f_*\mathcal{O}_X^\chi = \sum_\chi d_\chi^2 = n,$$

where χ runs over all irreducible characters of G .

We conclude this section with a lemma we will use in Chapter 2.

Lemma 1.17 *Let $f: X \rightarrow Y$ be a Galois covering of curves with Galois group G and $\text{char } k \nmid \#G$. Let χ be an irreducible character of G . Then the functor*

$$\mathcal{F} \mapsto (f_*\mathcal{F})^\chi$$

from the category of quasi-coherent $\mathcal{O}_X$ -modules to the category of quasi-coherent $\mathcal{O}_Y$ -modules is exact.

Proof. Let $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$ be a short exact sequence on X . By Lemma 1.4, the sequence $0 \rightarrow f_*\mathcal{F}_1 \rightarrow f_*\mathcal{F}_2 \rightarrow f_*\mathcal{F}_3 \rightarrow 0$ is again exact. Now $\mathcal{F}_i$ is a sheaf of $\mathcal{O}_Y[G]$ -modules, and $\mathcal{O}_Y[G] = \mathcal{O}_Y \otimes_k k[G] = \prod_\tau (\mathcal{O}_Y \otimes_k k[G]^\tau)$, where τ runs over all irreducible characters of G . Correspondingly, each $f_*\mathcal{F}_i$ decomposes as a direct sum of sheaves $(f_*\mathcal{F}_i)^\tau$, and the maps $f_*\mathcal{F}_i \rightarrow f_*\mathcal{F}_{i+1}$ decompose as a direct sum of maps $(f_*\mathcal{F}_i)^\tau \rightarrow (f_*\mathcal{F}_{i+1})^\tau$. From this the lemma readily follows. $\square$

Remark Instead of writing $(f_*\mathcal{F})^\chi$, we often just write $f_*\mathcal{F}^\chi$ to avoid messy formulas with an overkill of brackets.

1.4 Local monodromy

Let $f: X \rightarrow Y$ be a Galois covering with group G and branch points $\{Q_1, \dots, Q_r\}$. Let n be the order of G (assumed to be prime to the characteristic of the ground field), and fix a primitive n -th root of unity $\zeta_n \in k$. If e is a divisor of n , define $\zeta_e := (\zeta_n)^{n/e}$.

For each of the branch points Q_μ , choose a point $P_\mu \in X$ with $f(P_\mu) = Q_\mu$. The ramification index of P_μ is independent of the choice of P_μ in the fibre of Q_μ , call it e_μ . By Proposition 1.11 the stabilizer subgroup $G_{P_\mu} \subset G$ is cyclic of order e_μ . For each $\mu \in \{1, \dots, r\}$, let $\eta_\mu: G_{P_\mu} \rightarrow k^*$ be the primitive character of order e_μ . For each $\mu \in \{1, \dots, r\}$, let $\gamma_\mu \in G_{P_\mu}$ be the unique element with $\eta_\mu(\gamma_\mu) = \zeta_{e_\mu}$. It is clear that γ_μ is a generator of G_{P_μ} ; we call this element the *local monodromy* around Q_μ .

The element γ_μ obtained in this way depends on the choice of the point P_μ above Q_μ . Any other point above Q_μ is of the form $P'_\mu = \sigma \cdot P_\mu$ for some $\sigma \in G$. We then have $G_{P'_\mu} = \sigma G_{P_\mu} \sigma^{-1}$, and the fundamental character $\eta'_\mu: G_{P'_\mu} \rightarrow k^*$ at the point P'_μ is related to η_μ by $\eta'_\mu(\sigma g \sigma^{-1}) = \eta_\mu(g)$. Hence if we choose P'_μ as the point above Q_μ the local monodromy element that we obtain is $\gamma'_\mu = \sigma \gamma_\mu \sigma^{-1}$. In particular we see that the conjugacy class of γ_μ is independent of choices.

Let us briefly explain the connection of the above with the fundamental group of the open curve $Y' := Y \setminus \{Q_1, \dots, Q_r\}$. We first do this for the ground field $\mathbb{C}$, and we view all varieties involved as complex analytic varieties.

If g is the genus of Y and $y_0 \in Y'$ is a base point, it is known that the fundamental group $\pi_1(Y', y_0)$ has a presentation

$$\pi_1(Y', y_0) = \langle a_1, \dots, a_g, b_1, \dots, b_g, c_1, \dots, c_r \mid \prod_{i=1}^g [a_i, b_i] \cdot c_1 \cdots c_r = 1 \rangle,$$

that is, $\pi_1(Y', y_0)$ is the free group on generators $a_1, \dots, a_g, b_1, \dots, b_g, c_1, \dots, c_r$ with relation $\prod_{i=1}^g [a_i, b_i] \cdot c_1 \cdots c_r = 1$. Here the element c_μ is represented by a positively oriented loop around Q_μ . This presentation of the fundamental group is not canonical.

The Galois covering $f: X \rightarrow Y$ corresponds to a surjective homomorphism $\Phi: \pi_1(Y', y_0) \twoheadrightarrow G$, and with a suitable choice of the above presentation of the fundamental group we have that $\gamma_\mu = \Phi(c_\mu)$. For later application in examples, let us remark that for $Y \cong \mathbb{P}^1$ this implies the relation $\prod \gamma_\mu = 1$.

Over an arbitrary (algebraically closed) ground field, we no longer can apply the usual theory of the fundamental group. In its place comes Grothendieck's theory of the algebraic fundamental group, which, however, is beyond the scope of this thesis.

Consider a representation $\rho: G \rightarrow \mathrm{GL}(V)$ of degree d . Choose an index $\mu \in \{1, \dots, r\}$ and let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of $\rho(\gamma_\mu)$, taken with multiplicities. Since γ_μ has order e_μ , these eigenvalues are e_μ -th roots of unity. So with $\zeta_{e_\mu} = \zeta_n^{n/e_\mu}$ as above, we can write $\lambda_i = \zeta_{e_\mu}^{\alpha_i}$ for some $\alpha_i \in \{0, 1, \dots, e_\mu - 1\}$. For $\alpha \in \{0, \dots, e_\mu - 1\}$, let $N_{\mu\alpha}$ denote the number of eigenvalues λ_i that are equal to $\zeta_{e_\mu}^\alpha$. As remarked above, the conjugacy class of γ_μ is independent of choices; hence also the collection of eigenvalues λ_i and the numbers $N_{\mu,\alpha}$ do not depend on the choice of $P_\mu \in f^{-1}(Q_\mu)$.

Chapter 2

Proof of the Chevalley-Weil Formula

We recall the conventions we made in the previous chapter. The field k is algebraically closed and curves over k are assumed to be nonsingular and complete. Let X and Y be curves over k and let

$$f: X \rightarrow Y \tag{2.1}$$

be a Galois covering of degree n , with Galois group $G = \text{Cov}(X/Y)$. We assume that $\text{char } k \nmid \#G$.

Let $\{Q_1, \dots, Q_r\} \subseteq Y$ be the set of branch points of the morphism f and denote the local monodromy at Q_μ by $\gamma_\mu \in G$, as we defined precisely in §1.4. Write e_μ for the order of γ_μ in G , which is equal to the ramification index of f at a point lying above Q_μ , as we saw in Prop. 1.11.

2.1 The Chevalley-Weil formula

Let $f: X \rightarrow Y$ be a Galois covering as defined in equation 2.1. As we saw in §1.3, its Galois group $G = \text{Cov}(X/Y)$ has a left action on the vector space of global regular differential forms $H^0(X, \Omega_X^1)$ via $\sigma * \omega := (\sigma^{-1})^* \omega$, where $\sigma \in G$ and ω is a regular 1-form on X . This means we have a linear representation

$$G \rightarrow \text{GL}(H^0(X, \Omega_X^1))$$

of the group G .

The main question in this thesis is how many times a given irreducible representation of G occurs in $H^0(X, \Omega_X^1)$. If ρ_χ is an irreducible representation of G with character χ , then we denote this multiplicity by ν_χ . If we write d_χ for the degree of ρ_χ , we may reformulate our question as how to calculate

$$\nu_\chi d_\chi = \dim_k H^0(X, \Omega_X^1)^\chi.$$

Let γ_μ an element order e_μ in G of that represents the local monodromy of f at the branch point Q_μ and let $N_{\mu\alpha}$ denote the number of eigenvalues of $\rho_\chi(\gamma_\mu)$ that are equal to $\zeta_{e_\mu}^\alpha$, a power of a primitive e_μ -th root of unity, as we defined precisely in §1.4.

The following theorem is known as the Chevalley-Weil formula, which gives the answer to the above question. We write $\langle q \rangle = q - [q]$ for the *fractional part* of a rational number q . Be aware of $\langle -\frac{3}{7} \rangle = \frac{4}{7}$.

Theorem 2.1 (Chevalley-Weil, 1934) *Let $f: X \rightarrow Y$ be a Galois covering of curves with Galois group G . Assume $\text{char } k \nmid \#G$ and let the symbols e_μ and $N_{\mu\alpha}$ be defined as above. Then the multiplicity ν_χ of a given irreducible character χ in $H^0(X, \Omega_X^1)$ is given by*

$$\nu_\chi = d_\chi(g_Y - 1) + \sum_{\mu=1}^r \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \langle -\frac{\alpha}{e_\mu} \rangle + \sigma,$$

where $\sigma = 1$ if χ is the trivial character and $\sigma = 0$ otherwise.

As a special case, note that if χ is the trivial character then the Chevalley-Weil formula gives $\nu_\chi = g_Y$. In fact, we shall see in Corollary 2.7 that the map $f^*: H^0(Y, \Omega_Y^1) \rightarrow H^0(X, \Omega_X^1)^G$ is an isomorphism.

Example An example will enlighten all of this mysterious machinery. In Section 2.5 we will analyse two Galois coverings of the projective line.

2.2 Relative differentials

In this section we give some properties of the sheaf of relative differentials, which lead to a proof of the Riemann-Hurwitz formula at the end of this section. We include a proof of this well-known formula, since it has a lot in common with the proof of the Chevalley-Weil formula.

Proposition 2.2 *Let $f: X \rightarrow Y$ be a finite separable morphism of curves. Then there is a short exact sequence of sheaves on X ,*

$$0 \rightarrow f^* \Omega_Y^1 \rightarrow \Omega_X^1 \rightarrow \Omega_{X/Y}^1 \rightarrow 0.$$

Proof. Hartshorne [3], Prop. IV 2.1. □

The following proposition is in fact a weak version of Hartshorne [3], Prop. IV 2.2, but the proof given here prepares some tools that we will use in a later stage.

Proposition 2.3 *Let $f: X \rightarrow Y$ be a finite separable morphism of curves. Then*

- (i) *the support of $\Omega_{X/Y}^1$ is equal to the set of ramification points of f ;*
- (ii) *if P is a tamely ramified point on X with ramification index e_P , then the dimension of $\Omega_{X/Y, P}^1$, viewed as a k -vector space via the natural map $k \hookrightarrow \mathcal{O}_{X, P}$, is equal to $(e_P - 1)$.*

Proof. Take a point P on X and set $Q = f(P)$. Note that $(f^*\Omega_Y^1)_P = \Omega_{Y,Q}^1 \otimes_{\mathcal{O}_{Y,Q}} \mathcal{O}_{X,P}$. Choose uniformizers $t \in \mathcal{O}_{Y,Q}$ and $z \in \mathcal{O}_{X,P}$. The map $f^\#: \mathcal{O}_{Y,Q} \rightarrow \mathcal{O}_{X,P}$ is given by $t = uz^{e_P}$ for some unit $u \in \mathcal{O}_{X,P}^*$, with e_P the ramification index of f at P . Then $f^*: \Omega_{Y,Q}^1 \rightarrow \Omega_{X,P}^1$ sends dt to $d(uz^{e_P}) = (u'z + e_P u)z^{e_P-1}dz$. If P is not a ramification point then $e_P = 1$ and because $(u'z + u)$ is again a unit of $\mathcal{O}_{X,P}$ we see that the map $(f^*\Omega_Y^1)_P \rightarrow \Omega_{X,P}^1$ is an isomorphism. Hence by the previous proposition, $\Omega_{X/Y,P}^1 = 0$. If P is ramified then $e_P > 1$ and we see that the image of $(f^*\Omega_Y^1)_P \rightarrow \Omega_{X,P}^1$ is contained in $\mathfrak{m}_P \cdot \Omega_{X,P}^1$, so by the previous proposition we have $\Omega_{X/Y,P}^1 \neq 0$.

For (ii) suppose P is a tamely ramified point, which by definition means that $\text{char } k \nmid e_P$. Then $(u'z + e_P u)$ is a unit of $\mathcal{O}_{X,P}$, so we find that the image of the map $(f^*\Omega_Y^1)_P \rightarrow \Omega_{X,P}^1 = \mathcal{O}_{X,P} \cdot dz$ is the $\mathcal{O}_{X,P}$ -submodule $\mathcal{O}_{X,P} \cdot z^{e_P-1}dz = \mathfrak{m}_P^{e_P-1}dz$. Hence $\Omega_{X/Y,P}^1 \cong \mathcal{O}_{X,P}dz / \mathfrak{m}_P^{e_P-1}dz$, which has dimension $e_P - 1$ over $k \subset \mathcal{O}_{X,P}$. A basis is given by $dz, zdz, \dots, z^{e_P-2}dz$. $\square$

Remark Due to Lemma 1.3, we may notice that if f has only tame ramification, the dimension of $H^0(X, \Omega_{X/Y}^1)$ is equal to the total branching order of f , since

$$\dim_k H^0(X, \Omega_{X/Y}^1) = \dim_k \bigoplus_{P \text{ ramified}} \Omega_{X/Y,P}^1 = \sum_{P \text{ ramified}} e_P - 1.$$

Theorem 2.4 (Riemann-Hurwitz Formula) *Let $f: X \rightarrow Y$ be a finite separable morphism of curves with only tame ramification. Let b denote the total branching order of f and n the degree of f . Then*

$$g_X = n(g_Y - 1) + \frac{1}{2}b + 1.$$

Proof. We start with the exact sequence of Proposition 2.2 and apply the functor $\mathcal{F} \mapsto f_*\mathcal{F}$, which is exact by Lemma 1.4. Due to the Projection formula (Hartshorne [3], Ex. II 5.1), we end up with the following exact sequence of sheaves on Y ,

$$0 \rightarrow f_*\mathcal{O}_X \otimes_{\mathcal{O}_Y} \Omega_Y^1 \rightarrow f_*\Omega_X^1 \rightarrow f_*\Omega_{X/Y}^1 \rightarrow 0.$$

If we write down the long cohomology sequence and take dimensions, then we get the relation $\text{EC}(Y, f_*\mathcal{O}_X \otimes_{\mathcal{O}_Y} \Omega_Y^1) - \text{EC}(Y, f_*\Omega_X^1) + \text{EC}(Y, f_*\Omega_{X/Y}^1) = 0$, where EC denotes the Euler Characteristic; see §1.1. By Lemma 1.5, we have

$$\text{EC}(Y, f_*\Omega_X^1) = \text{EC}(X, \Omega_X^1) = g_X - 1$$

and according to Proposition 2.3 and Lemma 1.3, we have

$$\text{EC}(Y, f_*\Omega_{X/Y}^1) = \text{EC}(X, \Omega_{X/Y}^1) = b.$$

The sheaf $f_*\mathcal{O}_X \otimes_{\mathcal{O}_Y} \Omega_Y^1$ is a locally free $\mathcal{O}_Y$ -module of rank equal to n , since Ω_Y^1 is an invertible sheaf and $f_*\mathcal{O}_X$ is locally free of rank n by Proposition 1.16.

By Riemann-Roch (Theorem 1.2), we find

$$\begin{aligned}
\mathrm{EC}(Y, f_* \mathcal{O}_Y^1 \otimes_{\mathcal{O}_Y} \Omega_Y^1) &= n(1 - g_Y) + \int_Y c_1(f_* \mathcal{O}_Y^1 \otimes_{\mathcal{O}_Y} \Omega_Y^1) \\
&= n(1 - g_Y) + n \int_Y c_1(\Omega_Y^1) + \int_Y c_1(f_* \mathcal{O}_X) \\
&= n(1 - g_Y) + n(2g_Y - 2) + \{\mathrm{EC}(Y, f_* \mathcal{O}_X) - n(1 - g_Y)\} \\
&= n(2g_Y - 2) + 1 - g_X.
\end{aligned}$$

If we substitute this in the sum of Euler characteristics mentioned above and divide both sides by two, then we get the desired formula. $\square$

2.3 Projection on isotypic components

Let ρ_χ be an irreducible representation of G with character χ and degree d_χ . In order to calculate the dimension of $H^0(X, \Omega_X^1)^\chi$, we start with the exact sequence of Proposition 2.2 and look at the χ -isotypic components of the sheaves. Remember that we just write $f_* \mathcal{O}_X^\chi$ instead of $(f_* \mathcal{O}_X)^\chi$ to avoid messy formulas with an overkill of brackets.

Lemma 2.5 *Let $f: X \rightarrow Y$ be a Galois covering of curves with group G , $\mathrm{char} k \nmid \#G$, and let χ be an irreducible character of G . Then we have the following exact sequence of sheaves on Y ,*

$$0 \rightarrow f_* \mathcal{O}_X^\chi \otimes_{\mathcal{O}_Y} \Omega_Y^1 \rightarrow (f_* \Omega_X^1)^\chi \rightarrow (f_* \Omega_{X/Y}^1)^\chi \rightarrow 0.$$

Proof. This follows almost immediately from the exact sequence of Proposition 2.2. We apply the functor $\mathcal{F} \mapsto f_* \mathcal{F}^\chi$, which is exact by Lemma 1.17. If we use the Projection formula ([3] Ex. II 5.1), then the only thing we have to show is that

$$(f_* \mathcal{O}_X \otimes_{\mathcal{O}_Y} \Omega_Y^1)^\chi \cong f_* \mathcal{O}_X^\chi \otimes \Omega_Y^1.$$

This is clear, since G acts trivially on $\mathcal{O}_Y$ and Ω_Y^1 . $\square$

Let us recall some facts from Section 1.4. The morphism f is branched at the points $\{Q_1, \dots, Q_r\} \subseteq Y$ and we denoted the local monodromy at Q_μ by the element $\gamma_\mu \in G$, which is of order e_μ . The linear map $\rho_\chi(\gamma_\mu)$ has eigenvalues $\{\lambda_1, \dots, \lambda_{d_\chi}\}$ in k^* , which are powers of ζ_{e_μ} . We write $N_{\mu\alpha}$ for the number of eigenvalues of $\rho_\chi(\gamma_\mu)$ equal to $\zeta_{e_\mu}^\alpha$ for $\alpha = 0, \dots, e_\mu - 1$.

The following lemma is the first big step in proving the Chevalley-Weil formula.

Lemma 2.6 *Let $f: X \rightarrow Y$ be a Galois covering of curves with Galois group G . Assume $\mathrm{char} k \nmid \#G$ and let the symbols e_μ and $N_{\mu\alpha}$ be defined as in Section 1.4, mentioned above. Then*

$$\dim_k H^0(X, \Omega_{X/Y}^1)^\chi = d_\chi \sum_{\mu=1}^r \sum_{\alpha=1}^{e_\mu-1} N_{\mu\alpha}.$$

Proof. By Proposition 2.3 the cohomology of $\Omega_{X/Y}^1$ in degree 0 is a direct sum of all its stalks at ramification points. Since X/Y is Galois, we have for two points $P, P' \in f^{-1}(Q)$ that $\Omega_{X/Y,P}^1 \cong \Omega_{X/Y,P'}^1$. Therefore we may write

$$H^0(X, \Omega_{X/Y}^1)^X = \left[\bigoplus_{P \text{ ramified}} \Omega_{X/Y,P}^1 \right]^X \cong \bigoplus_{\mu=1}^r \left[\bigoplus_{P \in f^{-1}(Q_\mu)} \Omega_{X/Y,P}^1 \right]^X.$$

Fix some Q_μ and let us simply write $Q = Q_\mu$. Choose a point $P \in f^{-1}(Q)$. Since X/Y is Galois, the fibre $f^{-1}(Q)$ is in bijection with G/G_P , and for $\sigma \in G$ we have an isomorphism

$$(\sigma^{-1})^*: \Omega_{X/Y,P}^1 \xrightarrow{\sim} \Omega_{X/Y,\sigma P}^1.$$

We find that

$$\bigoplus_{P \in f^{-1}(Q)} \Omega_{X/Y,P}^1 = \bigoplus_{\sigma \in G/G_P} \Omega_{X/Y,\sigma P}^1 \cong \text{Ind}_{G_P}^G \Omega_{X/Y,P}^1.$$

The dimension of the χ -isotypic component is equal to the dimension of χ times the multiplicity of χ . By the Frobenius reciprocity formula (Prop. 1.7), we get

$$\dim_k \left[\text{Ind}_{G_P}^G \Omega_{X/Y,P}^1 \right]^\chi = d_\chi \left\langle \chi, \text{Ind}_{G_P}^G \Omega_{X/Y,P}^1 \right\rangle_G = d_\chi \left\langle \text{Res}_{G_P}^G \chi, \Omega_{X/Y,P}^1 \right\rangle_{G_P}.$$

Denote the character of G_P acting on the Zariski cotangent space at P by η . According to Proposition 1.15, we can choose uniformizers $t \in \mathcal{O}_{Y,Q}$ and $z \in \mathcal{O}_{X,P}^\eta$ such that $t \mapsto z^{e_P}$. Then $\mathcal{O}_{X,P}$ is free of rank e_P as a module over $\mathcal{O}_{X,Q}$ and $\{1, z, z^2, \dots, z^{e_P-1}\}$ is a basis. Further we have seen in (the proof of) Proposition 2.3 that $\Omega_{X/Y,P}^1$ has dimension $e_P - 1$ as a k -vector space and that $\{dz, z dz, z^2 dz, \dots, z^{e_P-2} dz\}$ is a basis. Since z is in the η -eigenspace of $\mathcal{O}_{X,P}$, we see that an element τ of G_P acts on a basis element $z^i dz$ by

$$\tau * (z^i dz) = (\eta(\tau)z)^i d(\eta(\tau)z) = \eta^{i+1}(\tau) \cdot z^i dz.$$

Thus the k -subspace of $\Omega_{X/Y,P}^1$ spanned by $z^i dz$ has character η^{i+1} for $i = 0, \dots, e_P - 2$. So every irreducible character of G_P occurs precisely once, except for the trivial character. Since $N_{\mu\alpha}$ is equal to the multiplicity of η^α in $\text{Res}_{G_P}^G \chi$, we get

$$\left\langle \text{Res}_{G_P}^G \chi, \Omega_{X/Y,P}^1 \right\rangle_{G_P} = \left[\begin{array}{c} \text{Number of nontrivial} \\ \text{eigenvalues of } \rho_\chi(\gamma_\mu) \end{array} \right] = \sum_{\alpha=1}^{d_\chi} N_{\mu\alpha}.$$

Summing over all $\mu = 1, \dots, r$ gives the desired result. $\square$

Corollary 2.7 *Let $f: X \rightarrow Y$ be a Galois covering of curves with Galois group G . Assume $\text{char } k \nmid \#G$. Then the subspace of G -invariant 1-forms in $H^0(X, \Omega_X^1)$ has dimension g_Y , the genus of the curve Y . Furthermore, G acts trivially on the space $H^1(X, \Omega_X^1)$.*

Proof. We take the cohomology sequence obtained from the sequence of Corollary 2.5, where we choose χ to be the trivial representation:

$$\begin{aligned} 0 &\rightarrow H^0(Y, f_* \mathcal{O}_X^G \otimes_{\mathcal{O}_Y} \Omega_Y^1) \rightarrow H^0(X, \Omega_X^1)^G \rightarrow H^0(X, \Omega_{X/Y}^1)^G \rightarrow \\ &\rightarrow H^1(Y, f_* \mathcal{O}_X^G \otimes_{\mathcal{O}_Y} \Omega_Y^1) \rightarrow H^1(X, \Omega_X^1)^G \rightarrow 0 \end{aligned}$$

By Proposition 1.16, we have $f_* \mathcal{O}_X^G \cong \mathcal{O}_Y$, so that we can replace $f_* \mathcal{O}_X^G \otimes_{\mathcal{O}_Y} \Omega_Y^1$ by Ω_Y^1 . We know from Lemma 2.6, that no stalk of $\Omega_{X/Y}^1$ contains a nontrivial subspace that is invariant under G_P . It follows immediately that $H^0(X, \Omega_{X/Y}^1)^G = 0$, so that the exact sequence gives us two isomorphisms:

$$\begin{aligned} H^0(X, \Omega_X^1)^G &\cong H^0(Y, \Omega_Y^1) \quad \text{which is } g_Y\text{-dimensional over } k, \\ H^1(X, \Omega_X^1)^G &\cong H^1(Y, \Omega_Y^1) \quad \text{which is 1-dimensional over } k. \end{aligned}$$

The last assertion now follows from the remark that $H^1(X, \Omega_X^1)$ is 1-dimensional over k , so we must have $H^1(X, \Omega_X^1)^G = H^1(X, \Omega_X^1)$ for dimension reasons. $\square$

Now we are ready to formulate a preliminary version of the Chevalley-Weil formula.

Proposition 2.8 (pre-Chevalley-Weil Formula) *Let $f: X \rightarrow Y$ be a Galois covering of curves with Galois group G . Assume $\text{char } k \nmid \#G$ and let the symbols e_μ and $N_{\mu\alpha}$ be as given in Section 1.4. Then the multiplicity ν_χ of an irreducible character χ in $H^0(X, \Omega_X^1)$ is given by*

$$\nu_\chi = d_\chi(g_Y - 1) + \frac{1}{d_\chi} \int_Y c_1(f_* \mathcal{O}_X^\chi) + \sum_{\mu=1}^r \sum_{\alpha=1}^{e_\mu-1} N_{\mu\alpha} + \sigma,$$

where $\sigma = 0$ unless χ is the trivial representation. In that case we have $\sigma = 1$ and thus $\nu_0 = g_Y$.

Proof. The exact sequence of Corollary 2.5 gives the following exact cohomology sequence,

$$\begin{aligned} 0 &\rightarrow H^0(Y, f_* \mathcal{O}_X^\chi \otimes_{\mathcal{O}_Y} \Omega_Y^1) \rightarrow H^0(X, \Omega_X^1)^\chi \rightarrow H^0(X, \Omega_{X/Y}^1)^\chi \rightarrow \\ &\rightarrow H^1(Y, f_* \mathcal{O}_X^\chi \otimes_{\mathcal{O}_Y} \Omega_Y^1) \rightarrow H^1(X, \Omega_X^1)^\chi \rightarrow 0 \end{aligned}$$

Now we take dimensions. Note that we want to calculate $\nu_\chi d_\chi = \dim_k H^0(X, \Omega_X^1)^\chi$. By Corollary 2.7, we may define

$$\sigma = \dim_k H^1(X, \Omega_X^1)^\chi = \begin{cases} 1 & \text{if } \chi \text{ is trivial,} \\ 0 & \text{otherwise.} \end{cases}$$

With use of Lemma 2.6, we have the following equation,

$$\text{EC}(Y, f_* \mathcal{O}_X^\chi \otimes_{\mathcal{O}_Y} \Omega_Y^1) - \nu_\chi d_\chi + d_\chi \sum_{\mu=1}^r \sum_{\alpha=1}^{e_\mu-1} N_{\mu\alpha} + \sigma = 0.$$

Since Ω_Y^1 is an invertible sheaf on Y and by Proposition 1.16 the sheaf $f_*\mathcal{O}_X^\chi$ is locally free of rank d_χ^2 , its tensor product is also a locally free $\mathcal{O}_Y$ -module of rank d_χ^2 . By Riemann-Roch (Theorem 1.2) we have

$$\begin{aligned} \text{EC}(Y, f_*\mathcal{O}_X^\chi \otimes_{\mathcal{O}_Y} \Omega_Y^1) &= d_\chi^2(1 - g_Y) + \int_Y c_1(f_*\mathcal{O}_X^\chi \otimes_{\mathcal{O}_Y} \Omega_Y^1) \\ &= d_\chi^2(1 - g_Y) + d_\chi^2 \int_Y c_1(\Omega_Y^1) + c_1(f_*\mathcal{O}_X^\chi) \\ &= d_\chi^2(g_Y - 1) + \int_Y c_1(f_*\mathcal{O}_X^\chi). \end{aligned}$$

The result follows after substitution and dividing both sides by d_χ . Note that we always have $\sigma/d_\chi = \sigma$, since we have that either σ equals zero, or $d_\chi = 1$, when χ is the trivial representation. $\square$

2.4 The fibred product

In order to compute the degree of $f_*\mathcal{O}_X^\chi$, we consider the fibred product of X with itself over Y . Recall that we may write this product, as a set, as

$$X \times_Y X = \{(P_1, P_2) \mid P_i \in X, f(P_1) = f(P_2)\}.$$

We are given projection morphisms, pr_1 and pr_2 , from $X \times_Y X$ onto X , on the first, respectively second coordinate.

Now consider the variety $G \times X$, which is a disjoint collection of n identical curves, together with the morphisms

$$\begin{array}{llll} G \times X & \rightarrow & X, & (\sigma, P) \mapsto \sigma P \quad \text{group action,} \\ G \times X & \rightarrow & X, & (\sigma, P) \mapsto P \quad \text{projection on 2nd coordinate.} \end{array}$$

Then, by definition of the fibred product (see Hartshorne [3], Thm. II 3.3), there exists a unique morphism

$$\psi: G \times X \rightarrow X \times_Y X,$$

such that the diagram of Figure 2.1 commutes.

Remark We use p_2 for the second projection map $G \times X \rightarrow X$ in order to avoid confusion with the projection map pr_2 . Note that $\text{pr}_2 \circ \psi = p_2$.

It is clear that the morphism ψ sends (σ, P) to $(\sigma P, P)$. Because the covering X/Y is Galois, this map is surjective. It follows that the corresponding structure sheaf homomorphism $\psi^\#$ is injective, so that we obtain a short exact sequence of sheaves on $X \times_Y X$,

$$0 \rightarrow \mathcal{O}_{X \times_Y X} \rightarrow \psi_*\mathcal{O}_{G \times X} \rightarrow \mathcal{Q} \rightarrow 0, \quad (2.2)$$

where we simply define $\mathcal{Q}$ as the quotient sheaf.

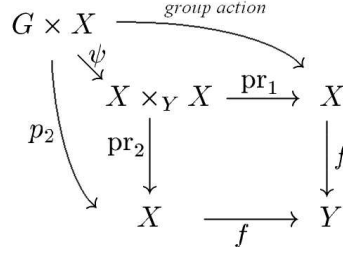


Figure 2.1: The fibred product $X \times_Y X$ compared to the variety $G \times X$

Because the covering $f: X \rightarrow Y$ is finite, the projection morphism $\text{pr}_2: X \times_Y X \rightarrow X$ is finite as well, of the same degree as f . By push forward we therefore get a short exact sequence of $\mathcal{O}_X$ -modules,

$$0 \rightarrow \text{pr}_{2,*} \mathcal{O}_{X \times_Y X} \rightarrow p_{2,*} \mathcal{O}_{G \times X} \rightarrow \text{pr}_{2,*} \mathcal{Q} \rightarrow 0. \quad (2.3)$$

We let the group G act on $X \times_Y X$ by its action on the first factor, so $\sigma \cdot (P_1, P_2) = (\sigma P_1, P_2)$. Note that this is an action over X , where we view $X \times_Y X$ as a scheme over X via the second projection. We let G act on $G \times X$ via left translations in the group; so $\sigma \cdot (g, P) = (\sigma g, P)$. Again this is an action over X , where we view $G \times X$ as a scheme over X via the second projection. With these G -actions the morphism ψ is G -equivariant. It follows that the sheaf homomorphism $\text{pr}_{2,*} \mathcal{O}_{X \times_Y X} \rightarrow p_{2,*} \mathcal{O}_{G \times X}$ is G -equivariant too. Hence we obtain a natural G -action on $\text{pr}_{2,*} \mathcal{Q}$.

Thus the group G acts on each sheaf in the exact sequence (2.3) and due to Lemma 1.17, we may look at the χ -isotypic components,

$$0 \rightarrow \text{pr}_{2,*} \mathcal{O}_{X \times_Y X}^\chi \rightarrow p_{2,*} \mathcal{O}_{G \times X}^\chi \rightarrow \text{pr}_{2,*} \mathcal{Q}^\chi \rightarrow 0.$$

Back to where we started. Remember that we wanted to compute the degree of $f_* \mathcal{O}_X^\chi$. The following lemma motivates why we introduced the fibred product.

Lemma 2.9 *Let $f: X \rightarrow Y$ be a Galois covering of curves. Assume that $\text{char } k \nmid \#G$. Let χ be an irreducible character of G . Then we have the equality*

$$\int_X c_1(f_* \mathcal{O}_X^\chi) = -\frac{1}{\deg(f)} \int_X c_1(\text{pr}_{2,*} \mathcal{Q}^\chi).$$

Proof. We apply Hartshorne [3], Prop. III 9.3 to the situation drawn in Figure 2.1. It follows that we have a natural isomorphism $f^* f_* \mathcal{O}_X \cong \text{pr}_{2,*} \text{pr}_1^* \mathcal{O}_X = \text{pr}_{2,*} \mathcal{O}_{X \times_Y X}$. It is clear from the definitions that this isomorphism is G -equivariant, so for any character χ it restricts to an isomorphism $f^*((f_* \mathcal{O}_X)^\chi) \cong (\text{pr}_{2,*} \mathcal{O}_{X \times_Y X})^\chi$. Taking degrees it follows from Lemma 1.1 that we have

$$\int_Y c_1(f_* \mathcal{O}_X^\chi) = \frac{1}{\deg(f)} \int_X c_1(\text{pr}_{2,*} \mathcal{O}_{X \times_Y X}^\chi).$$

Because of the short exact sequence 2.3, the degree of $\mathrm{pr}_{2,*}\mathcal{O}_{X \times_Y X}^\chi$ is equal to the degree of $p_{2,*}\mathcal{O}_{G \times X}^\chi$ minus the degree of $\mathrm{pr}_{2,*}\mathcal{Q}^\chi$ (cf. Hartshorne [3], Ex. II 6.12). So we only have to prove that $\int_X c_1(p_{2,*}\mathcal{O}_{G \times X}^\chi) = 0$.

Viewing G as a constant group scheme we have $G = \mathrm{Spec}(k^G)$, where $k^G = \{\text{maps } G \rightarrow k\}$. Hence $p_{2,*}\mathcal{O}_{G \times X} = \underline{k}^G \otimes_k \mathcal{O}_X$, where $\underline{k}^G$ denotes the constant sheaf associated to k^G . The action of G on $p_{2,*}\mathcal{O}_{G \times X}$ comes from its action on $\underline{k}^G$, thus we have

$$[p_{2,*}\mathcal{O}_{G \times X}]^\chi = [\underline{k}^G]^\chi \otimes_k \mathcal{O}_X.$$

As this is a free $\mathcal{O}_X$ -module its degree is zero. $\square$

Choose a point (P_1, P_2) on $X \times_Y X$ and let $\sigma \in G$ be such that $P_1 = \sigma P_2$. Take an open affine neighbourhood U_2 of P_2 in X and define $U_1 = \sigma U_2$, which is an open affine neighbourhood of P_1 in X . Write $V = f(U_1) = f(U_2)$. Then $U_1 \times_V U_2$ is an open neighbourhood of (P_1, P_2) and, by Hartshorne [3], Thm. II 3.3, we have

$$\mathcal{O}_{X \times_Y X}(U_1 \times_V U_2) \cong \mathcal{O}_X(U_1) \otimes_{\mathcal{O}_Y(V)} \mathcal{O}_X(U_2).$$

Let $\mathfrak{m}_{P_1}$ be the maximal ideal of $\mathcal{O}_X(U_1)$ corresponding to the point P_1 . Let $e = e_{P_1} = e_{P_2}$, and write $Q := f(P_1)$. Choose uniformizers $x \in \mathcal{O}_{X,P_1}$ and $t \in \mathcal{O}_{Y,Q}$ such that $t \mapsto x^e$, and let η be the associated character of G_{P_1} as in Prop. 1.15. Because $P_1 = \sigma P_2$, the element $y = \sigma^\# x \in \mathfrak{m}_{P_2}$ is a uniformizer of $\mathcal{O}_{X,P_2}$. Note that the homomorphism $f^\# : \mathcal{O}_{Y,Q} \rightarrow \mathcal{O}_{X,P_2}$ sends t to y^e .

Things become much more convenient when we take the completions with respect to the maximal ideal $\mathfrak{n}$ corresponding to the point (P_1, P_2) , which is $\mathfrak{n} = \mathfrak{m}_{P_1} \otimes \mathcal{O}_X(U_2) + \mathcal{O}_X(U_1) \otimes \mathfrak{m}_{P_2}$. Then we get

$$\hat{\mathcal{O}}_{X \times_Y X, (P_1, P_2)} \cong k[[x]] \hat{\otimes}_{k[[t]]} k[[y]] \cong k[[x, y]] / (x^e - y^e).$$

Note that when we write $\hat{\mathcal{O}}_{X \times_Y X, (P_1, P_2)} = k[[x, y]] / (x^e - y^e)$, formally speaking the “ x ” is $\mathrm{pr}_1^\#(x)$ and “ y ” is $\mathrm{pr}_2^\#(y)$.

In the exact sequence (2.2) on $X \times_Y X$ we take stalks at the point (P_1, P_2) , which gives an exact sequence of $\mathcal{O}_{X \times_Y X, (P_1, P_2)}$ -modules, and then take $\mathfrak{n}$ -adic completions, which is exact by Hartshorne [3], Thm. 9.3Ac. This gives an exact sequence

$$0 \rightarrow \hat{\mathcal{O}}_{X \times_Y X, (P_1, P_2)} \rightarrow \bigoplus_{g \in G_{P_1} \sigma} \hat{\mathcal{O}}_{\{g\} \times X, (g, P_2)} \rightarrow \hat{\mathcal{Q}}_{(P_1, P_2)} \rightarrow 0,$$

where g runs through $G_{P_1} \sigma$, i.e., we let g run through the set of all elements in G with $P_1 = gP_2$.

For $g \in G$, we restrict the morphism ψ to $\{g\} \times X$, so that we obtain a map $\psi_g : \{g\} \times X \rightarrow X \times_Y X$ that is given on points by $(g, P) \mapsto (gP, P)$. For $g \in G_{P_1} \sigma$ write $z^{(g)} := p_2^\#(y)$, which is a uniformizer of $\mathcal{O}_{\{g\} \times X, (g, P_2)}$. We can rewrite our exact sequence in the form

$$0 \rightarrow k[[x, y]] / (x^e - y^e) \xrightarrow{\psi^\#} \bigoplus_{g \in G_{P_1} \sigma} k[[z^{(g)}]] \rightarrow \hat{\mathcal{Q}}_{(P_1, P_2)} \rightarrow 0. \quad (2.4)$$

The homomorphism $\psi^\#$ in the sequence of 2.4 is in fact a collection of homomorphisms $(\psi_g^\#)_{g \in G_{P_1}\sigma}$ and sends an element $\phi \in k[[x, y]]/(x^e - y^e)$ to $(\psi_g^\#(\phi))_g$ in $\bigoplus_{g \in G_{P_1}\sigma} k[[z^{(g)}]]$. It follows readily from the definitions that $\psi_g^\#(y) = z^{(g)}$, since formally speaking “ y ” is $\text{pr}_2^\#(y)$ and thus we get $\psi_g^\# \text{pr}_2^\#(y) = p_2^\#(y) = z^{(g)}$.

Next we compute $\psi_g^\#(x)$, which formally is $\psi_g^\# \text{pr}_1^\#(x) = (\text{pr}_1 \circ \psi_g)^\#(x)$. Now $(\text{pr}_1 \circ \psi_g)$ is the map $\{g\} \times X \rightarrow X$ given by $(g, P) \mapsto gP$, and if we write $g = \gamma\sigma$ then this map equals the composition

$$\{g\} \times X \xrightarrow{p_2} X \xrightarrow{\sigma} X \xrightarrow{\gamma} X.$$

Because x was chosen to be an eigenvector for the fundamental character η we have $\gamma^\#(x) = \eta(\gamma^{-1})x$. Since by definition $y = \sigma^\#(x)$ and $z^{(g)} = p_2^\#(y)$ we then find that $\psi_g^\# \text{pr}_1^\#(x) = \eta(\gamma^{-1})z^{(g)}$.

We introduce some convenient notation. For $g \in G_{P_1}\sigma$, let $\mathbf{e}_g$ denote the identity element in $\hat{\mathcal{O}}_{\{g\} \times X, (g, P_2)}$. As we shall express elements of $\hat{\mathcal{O}}_{\{g\} \times X, (g, P_2)}$ in terms of this base vector $\mathbf{e}_g$ we can simply write z for each of the elements previously called $z^{(g)}$.

Now we can summarize the above abstract algebra by saying that $\psi^\#$ is the homomorphism from $k[[x, y]]/(x^e - y^e)$ to $\bigoplus_{g \in G_{P_1}\sigma} k[[z]]\mathbf{e}_g$, given by

$$\sum_{l=0}^{e-1} \sum_{m=0}^{\infty} \lambda_{lm} x^l y^m \mapsto \sum_{g \in G_{P_1}\sigma} \left(\sum_{l=0}^{e-1} \eta^l(\gamma^{-1}) \sum_{m=0}^{\infty} \lambda_{lm} z^{l+m} \right) \mathbf{e}_g.$$

The actions of the group G on $G \times X$ and $X \times_Y X$ give rise to a natural action of G_{P_1} on $\hat{\mathcal{Q}}_{(P_1, P_2)}$. Recall that G_{P_1} is cyclic of order e , and that η is a fundamental character of G_{P_1} .

Proposition 2.10 *Retaining the above notations we have*

$$\dim_k[\hat{\mathcal{Q}}_{(P_1, P_2)}]^{\eta^\alpha} = \alpha,$$

for $\alpha = 0, 1, \dots, e-1$.

Proof. The terms in the exact sequence (2.4) each have the structure of a module over $\hat{\mathcal{O}}_{X, P_2} = k[[y]]$ on which G_{P_1} acts, and the maps in the sequence are G_{P_1} -equivariant. The module $\hat{\mathcal{O}}_{X \times_Y X, (P_1, P_2)} = k[[x, y]]/(x^e - y^e)$ is free of rank e as a module over $k[[y]]$ with basis $\{1, x, \dots, x^{e-1}\}$ and G_{P_1} acts on x^α through the character η^α . For $\bigoplus_{g \in G_{P_1}\sigma} \hat{\mathcal{O}}_{\{g\} \times X, (g, P_2)} = \bigoplus_{g \in G_{P_1}\sigma} k[[z]] \cdot \mathbf{e}_g$, the $k[[y]]$ -module structure is the one obtained from the homomorphism of rings $k[[y]] \rightarrow \bigoplus_{g \in G_{P_1}\sigma} k[[z]] \cdot \mathbf{e}_g$ that sends y to $\sum z \mathbf{e}_g$, and $\delta \in G_{P_1}$ sends $\mathbf{e}_g$ to $\mathbf{e}_{\delta^{-1}g}$. Define elements $\mathbf{f}_\alpha$ for $\alpha = 0, \dots, e-1$ by

$$\mathbf{f}_\alpha := \sum_{\delta \in G_{P_1}} \eta^\alpha(\delta) \cdot (\delta * \mathbf{e}_\sigma) = \sum_{\gamma \in G_{P_1}} \eta^\alpha(\gamma^{-1}) \mathbf{e}_{\gamma\sigma}.$$

Then $\mathbf{f}_\alpha$ generates the $k[[y]]$ -submodule corresponding to the character η^α .

By our previous computations we know that the map $\psi^\#$ sends an element $\phi = \sum_{m=0}^{\infty} \lambda_m x^\alpha y^m$ in the η^α -eigenspace of $k[[x, y]]/(x^e - y^e)$ to

$$\sum_{\gamma \in G_{P_1}} \sum_{m=0}^{\infty} \lambda_m \eta^\alpha(\gamma^{-1}) z^{\alpha+m} \mathbf{e}_{\gamma\sigma} = \sum_{m=0}^{\infty} \lambda_m z^{\alpha+m} \cdot \mathbf{f}_\alpha.$$

From this it readily follows that $[\hat{\mathcal{Q}}_{(P_1, P_2)}]^\eta$ equals $k[[z]] \cdot \mathbf{f}_\alpha / k[[z]] \cdot z^\alpha \mathbf{f}_\alpha$; in particular it has dimension α as a k -vector space. $\square$

Corollary 2.11 *The support of the sheaf $\mathcal{Q}$ is the set of points $(P_1, P_2) \in X \times_Y X$ such that P_1 and P_2 are ramification points of the morphism f .*

We can now calculate the degrees of the sheaves $\mathrm{pr}_{2,*} \mathcal{Q}^\chi$. It follows from the previous Corollary that the sheaf $\mathrm{pr}_{2,*} \mathcal{Q}$ on X has support only at the ramification points of f , so according to Hartshorne [3], Ex. II 6.12, we may say that it is a *torsion sheaf* and we may calculate its degree as follows,

$$\int_X c_1(\mathrm{pr}_{2,*} \mathcal{Q}^\chi) = \sum_{P \text{ ramified} \in X} \dim_k(\mathrm{pr}_{2,*} \mathcal{Q})_P^\chi. \quad (2.5)$$

Hence it will suffice to calculate the dimensions of the stalks of $\mathrm{pr}_{2,*} \mathcal{Q}^\chi$. The result is as follows.

Proposition 2.12 *Let $f: X \rightarrow Y$ be a Galois covering of curves and consider the situation drawn in Figure 2.1. Assume that $\mathrm{char} k \nmid \#G$ and let $\mathcal{Q}$ be defined as in the exact sequence (2.2). Let $\{Q_1, \dots, Q_r\}$ denote the set of branch points of f and fix a ramification point P on X that is mapped to a given Q_μ . Let χ be an irreducible character of G . Then*

$$\dim_k(\mathrm{pr}_{2,*} \mathcal{Q})_P^\chi = d_\chi \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \alpha,$$

where e_μ and $N_{\mu\alpha}$ are defined as before in Section 1.4.

Proof. The fibre $\mathrm{pr}_2^{-1}(P)$ is in bijection with G/G_P by sending σG_P to $(\sigma P, P)$. We have

$$(\mathrm{pr}_{2,*} \mathcal{Q})_P = \bigoplus_{[\sigma] \in G/G_P} \mathcal{Q}_{(\sigma P, P)}.$$

An element $\gamma \in G_P$ acts on this via the collection of isomorphisms $(\gamma^{-1})^*: \mathcal{Q}_{(\sigma P, P)} \xrightarrow{\sim} \mathcal{Q}_{(\gamma\sigma P, P)}$. This just means that $(\mathrm{pr}_{2,*} \mathcal{Q})_P \cong \mathrm{Ind}_{G_P}^G \mathcal{Q}_{(P, P)}$. Now we apply the Frobenius reciprocity formula (Prop. 1.7) and Proposition 2.10. Since η^α occurs exactly $N_{\mu\alpha}$ times in $\mathrm{Res}_{G_P}^G \chi$, we find that

$$\begin{aligned} \dim_k \left[\mathrm{Ind}_{G_P}^G \mathcal{Q}_{(P, P)} \right]^\chi &= d_\chi \left\langle \chi, \mathrm{Ind}_{G_P}^G \mathcal{Q}_{(P, P)} \right\rangle_G = d_\chi \left\langle \mathrm{Res}_{G_P}^G \chi, \mathcal{Q}_{(P, P)} \right\rangle_{G_P} \\ &= d_\chi \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \alpha, \end{aligned}$$

which proves the corollary. $\square$

Now the Chevalley-Weil Theorem follows by putting it all together.

Theorem 2.13 (Chevalley-Weil, 1934) *Let $f: X \rightarrow Y$ be a Galois covering as given in Section 2.1. Assume that $\text{char } k \nmid \#G$. Let the symbols e_μ and $N_{\mu\alpha}$ be defined as before, in §1.4. Then the multiplicity ν_χ of an irreducible character χ in $H^0(X, \Omega_X^1)$ is given by*

$$\nu_\chi = d_\chi(g_Y - 1) + \sum_{\mu=1}^r \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \langle -\frac{\alpha}{e_\mu} \rangle + \sigma,$$

where $\sigma = 1$ if χ is the trivial character and $\sigma = 0$ otherwise.

Proof. By Proposition 2.8, we have the so-called pre-Chevalley-Weil formula,

$$\nu_\chi = d_\chi(g_Y - 1) + \frac{1}{d_\chi} \int_Y c_1(f_* \mathcal{O}_X^\chi) + \sum_{\mu=1}^r \sum_{\alpha=1}^{e_\mu-1} N_{\mu\alpha} + \sigma.$$

Let $n = \deg(f)$. By Lemma 2.9, equation (2.5) and Proposition 2.12, we find

$$\begin{aligned} \int_Y c_1(f_* \mathcal{O}_X^\chi) &= -\frac{1}{n} \sum_{P \text{ ramified}} \dim_k(\text{pr}_{2,*} \mathcal{Q})_P^\chi \\ &= -\frac{1}{n} \sum_{\mu=1}^r \sum_{P \in f^{-1}(Q_\mu)} d_\chi \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \alpha \end{aligned}$$

Since X/Y is Galois, we have that $\#f^{-1}(Q_\mu) = n/e_\mu$. It follows that

$$\int_Y c_1(f_* \mathcal{O}_X^\chi) = -\frac{1}{n} \sum_{\mu=1}^r \frac{n}{e_\mu} \left(d_\chi \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \alpha \right) = -d_\chi \sum_{\mu=1}^r \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \frac{\alpha}{e_\mu}.$$

Substituting this results in our pre-formula of Proposition 2.8 gives us our desired result:

$$\begin{aligned} \nu_\chi &= d_\chi(g_Y - 1) - \sum_{\mu=1}^r \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \frac{\alpha}{e_\mu} + \sum_{\mu=1}^r \sum_{\alpha=1}^{e_\mu-1} N_{\mu\alpha} + \sigma \\ &= d_\chi(g_Y - 1) + \sum_{\mu=1}^r \sum_{\alpha=1}^{e_\mu-1} N_{\mu\alpha} \left(1 - \frac{\alpha}{e_\mu}\right) + \sigma \\ &= d_\chi(g_Y - 1) + \sum_{\mu=1}^r \sum_{\alpha=0}^{e_\mu-1} N_{\mu\alpha} \langle -\frac{\alpha}{e_\mu} \rangle + \sigma \end{aligned}$$

Hurray, we have proven the Chevalley-Weil formula! □

2.5 Examples

To celebrate our achievement, we take a look at two concrete examples to see how this beautiful theory works.

A cyclic Galois cover of the projective line

Example Let us consider the curve X given by the equation

$$y^n = x(x-1)(x-\lambda)$$

and the morphism $f: X \rightarrow \mathbb{P}_k^1$ from X to the projective line, given by $(x, y) \mapsto x$, where we demand that $n \geq 4$ is not divisible by 3 and $\lambda \neq 0, 1, \infty$. We see that f has four branch points, with $x = 0, 1, \lambda, \infty$, which we denote by $\{Q_0, Q_1, Q_\lambda, Q_\infty\}$, all with branching order $e = n$. See Figure 2.2.

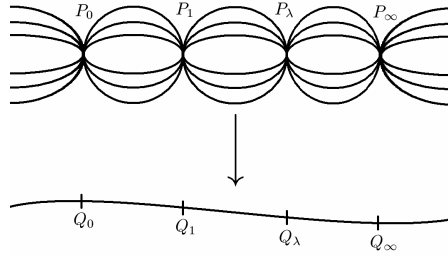


Figure 2.2: $y^n = x(x-1)(x-\lambda)$

Its Galois group G is isomorphic to $\mathbb{Z}/n\mathbb{Z}$. Because this group is cyclic, all its irreducible representations are powers of some fundamental representation η . An element σ acts on X as $(x, y) \mapsto (x, \eta(\sigma)y)$, i.e., if σ is a generator of G , then it is sent by η to ζ_n , a primitive n -th root of unity.

By Hurwitz, we find that X has genus $g_X = n-1$. In order to find the $n-1$ linearly independent regular 1-forms on all of X , we consider the divisors of the functions x and y and the one-form dx .

	P_0	P_1	P_λ	P_∞
(x)	n	0	0	$-n$
(y)	1	1	1	-3
(dx)	$n-1$	$n-1$	$n-1$	$-n-1$

We find that the 1-forms $\frac{dx}{y^i}$ with $i = \lceil \frac{n+1}{3} \rceil, \dots, n-1$ and $\frac{x dx}{y^i}$ with $i = \lceil \frac{2n+1}{3} \rceil, \dots, n-1$ do not have any pole on X . It is clear that these are $n-1$ linearly independent one-forms over k , so that they form a basis for $H^0(X, \Omega_X^1)$.

How does G act on these forms? We immediately find that

$$\sigma * \frac{x^j dx}{y^i} = \frac{x^j dx}{(\eta(\sigma)y)^i} = \eta^{-i}(\sigma) \frac{x^j dx}{y^i},$$

thus G acts via η^{-i} on the space spanned by $\frac{dx}{y^i}$ and $\frac{xdx}{y^i}$. Let us summarize our results in the following table.

0	$\leq i < \lceil \frac{n+1}{3} \rceil$	η^{-i} does not occur	$\eta^0, \eta^{\lceil \frac{2n-1}{3} \rceil+1}, \dots, \eta^{n-1}$
$\lceil \frac{n+1}{3} \rceil$	$\leq i < \lceil \frac{2n+1}{3} \rceil$	η^{-i} occurs once: $\frac{dx}{y^i}$	$\eta^{\lceil \frac{n-1}{3} \rceil+1}, \dots, \eta^{\lceil \frac{2n-1}{3} \rceil}$
$\lceil \frac{2n+1}{3} \rceil$	$\leq i \leq n-1$	η^{-i} occurs twice: $\frac{dx}{y^i}, \frac{xdx}{y^i}$	$\eta^1, \dots, \eta^{\lceil \frac{n-1}{3} \rceil}$

Let us check these results. The $\lceil \frac{n-2}{3} \rceil$ different representations that are mentioned in the second row all occur once in $H^0(X, \Omega_X^1)$. In the second row, we have $\lceil \frac{n-1}{3} \rceil$ different representations, but they occur twice. This makes a total of $n-1$ representations, which is indeed equal to g_X .

What do Chevalley and Weil say about his example? We have to look at the local monodromy at the branch points. We find that γ_0, γ_1 and γ_λ all are equal to σ , and that $\gamma_\infty = \sigma^{n-3}$.

The representation η^i of G sends elements σ^k of G to $(\eta(\sigma))^{ik}$. Therefore, we find that

$$\begin{array}{ll} \text{at } Q_0, Q_1, Q_\lambda & N_{\mu\alpha} = 1 \text{ for } \alpha = i \quad \text{and } N_{\mu\alpha} = 0 \text{ otherwise,} \\ \text{at } Q_\infty & N_{\mu\alpha} = 1 \text{ for } \alpha = (n-3)i \quad \text{and } N_{\mu\alpha} = 0 \text{ otherwise.} \end{array}$$

Then, according to the Chevalley-Weil formula, we find that the representation η^i occurs in $H^0(X, \Omega_X^1)$ with multiplicity

$$\nu_i = -1 + 3\langle -\frac{i}{n} \rangle + \langle -\frac{(n-3)i}{n} \rangle + \sigma = \begin{cases} 0 & \text{if } i = 0, \\ 2 - \frac{3i}{n} + \langle \frac{3i}{n} \rangle & \text{otherwise.} \end{cases}$$

We distinguish four cases. For the trivial representation, $i = 0$, we find $\nu_0 = 0$. For the representation η^i with $1 \leq i \leq \lfloor \frac{n}{3} \rfloor$, we find that $\langle \frac{3i}{n} \rangle = \frac{3i}{n}$, thus $\nu_i = 2$. For $\lfloor \frac{n}{3} \rfloor < i \leq \lfloor \frac{2n}{3} \rfloor$, we find $\langle \frac{3i}{n} \rangle = \frac{3i}{n} - 1$, thus $\nu_i = 1$. Similarly, we find $\nu_i = 0$ for $\lfloor \frac{2n}{3} \rfloor < i \leq n-1$.

Indeed, Chevalley and Weil give us the same answers as we just had found ourselves.

A dihedral Galois cover of the projective line

Example Consider the covering $f: X \rightarrow \mathbb{P}_k^1$, given by

$$y^n = \frac{x-x_1}{x+x_1} \frac{x-x_2}{x+x_2}, \quad (x, y) \mapsto x^2.$$

Its Galois group is the dihedral group D_n of $2n$ elements, which we present by the generators r and s and the relations $r^n = s^2 = (sr)^2 = 1$. This group acts on X by

$$r: (x, y) \rightsquigarrow (x, \zeta_n y), \quad s: (x, y) \rightsquigarrow (-x, \frac{1}{y}).$$

The morphism f has four branch points $\{Q_0, Q_{x_1^2}, Q_{x_2^2}, Q_\infty\}$, the corresponding local monodromy is given by $\{s, r, r, sr^2\}$. See Figure 2.3.

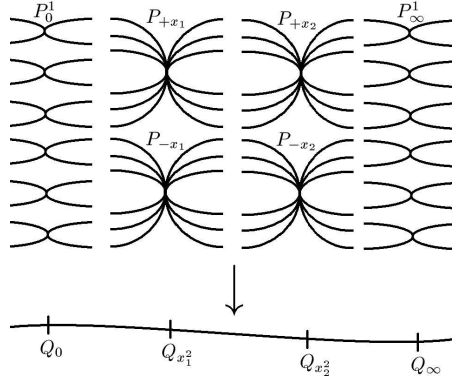


Figure 2.3: $y^n = \frac{x-x_1}{x+x_1} \frac{x-x_2}{x+x_2}$

By Hurwitz, we find that $g_X = -2n + \frac{1}{2}(2n + 4(n-1)) + 1 = n-1$. Let us write down the $n-1$ linearly independent regular 1-forms on all of X . We find them by considering some divisors on X .

	P_0^1	P_∞^1	P_{+x_1}	P_{-x_1}	P_{+x_2}	P_{-x_2}
(y)	0	0	1	-1	1	-1
$(x-x_1)$	0	-1	n	0	0	0
$(x-x_2)$	0	-1	0	0	n	0
(dx)	0	-2	$n-1$	$n-1$	$n-1$	$n-1$

We find that the 1-forms $\omega_\alpha = \frac{y^\alpha dx}{(x-x_1)(x-x_2)}$ with $\alpha = 1, \dots, n-1$ are regular on X . How does G act on these forms? We see that

$$r * \omega_\alpha = \frac{(\zeta_n y)^\alpha dx}{(x-x_1)(x-x_2)} = \zeta_n^\alpha \omega_\alpha, \quad s * \omega_\alpha = \frac{(1/y)^\alpha d(-x)}{(-x-x_1)(-x-x_2)} = -\omega_{n-\alpha}.$$

What are the irreducible representations of D_n ? We distinguish two cases, depending on whether n is even or odd. Motivation for the results can be found in Serre [5], §5.3. If n is even, then there exist four representations of degree 1, say $\psi_1, \dots, \psi_4$, and $\frac{n}{2} - 1$ representations of degree two, say ρ_h for $0 < h < n/2$. They are given by the following table.

	r^k	sr^k		r^k	sr^k
ψ_1	1	1	ρ_h	$\begin{pmatrix} \zeta_n^{hk} & 0 \\ 0 & \zeta_n^{-hk} \end{pmatrix}$	$\begin{pmatrix} 0 & \zeta_n^{-hk} \\ \zeta_n^{hk} & 0 \end{pmatrix}$
ψ_2	1	-1			
ψ_3	$(-1)^k$	$(-1)^k$			
ψ_4	$(-1)^k$	$(-1)^{k+1}$			

The case with odd n is not very different. Then there exist two one dimensional representations, the ψ_1 and ψ_2 written above, and $\frac{1}{2}(n-1)$ representations of degree two, the ρ_h with $0 < h < n/2$ written above.

Let us go back to our space of regular 1-forms. For each $n \geq 2$, every ρ_h occurs once and we write for its isotypical component

$$[H^0(X, \Omega_X^1)]^{\rho_h} = k \cdot \omega_h \oplus k \cdot (-\omega_{n-h}).$$

If n is odd, we see that this is the desired canonical decomposition of $H^0(X, \Omega_X^1)$, since $\frac{1}{2}(n-1) \times 2 = n-1 = g_X$. If n is even, then we find the representation ψ_4 as well, this is the space spanned by $\omega_{n/2}$.

What do Chevalley and Weil say about this example? Let us begin with the representation ρ_h for $0 < h < n/2$. The local monodromy around the branch points $Q_{x_1^2}$ and $Q_{x_2^2}$ is given by r and the eigenvalues of $\rho_h(r)$ are the roots ζ_n^h and ζ_n^{n-h} . Around Q_0 and Q_∞ , we have $\gamma_0 = s$ and $\gamma_\infty = sr^2$, and both corresponding matrices have eigenvalues 1 and -1 . Substituting these values in the Chevalley-Weil formula gives us for each $0 < h < n/2$,

$$\nu_h = -2 + 2(0 + \langle -\frac{1}{2} \rangle) + 2(\langle -\frac{n}{h} \rangle + \langle -\frac{n-h}{n} \rangle) = 1.$$

Similarly, we find that ν_{ψ_1} and ν_{ψ_2} are both equal to 0, and for the case that n is even, we find $\nu_{\psi_3} = 0$ and $\nu_{\psi_4} = 1$.

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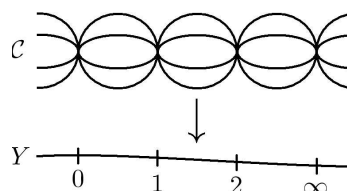
Furthermore, I want to mention the entire atmosphere in Euclides. It is really remarkable that a student can jump into a teacher's room with the request to study some particular subject, so that a new course is created; a course that is not in the *studiegids*. I did this twice, with Gerard van der Geer, about elliptic curves, and with Ben Moonen, mentioned above. In general, all teachers were very enthusiastic, always being patient to answer all the questions I had. Special attention goes to Iris Hettelingh, my *studieadviseur*.

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Samenvatting

voor de eerstejaars student

Bekijk de kromme $\mathcal{C}$, gegeven door de vergelijking $y^4 = x(x-1)(x-2)$, met x en y in $\mathbb{C}$. De kromme is afgebeeld in de figuur hiernaast, we zien dat hij symmetrisch is. We kunnen dit precies maken door de afbeelding $f: \mathcal{C} \rightarrow Y$ te definiëren, waarbij we met Y de x -as aangeven, en f een punt (x, y) op $\mathcal{C}$ afbeeldt op het punt x op Y , dus $f(x, y) = x$. Het heeft geen effect als we deze afbeelding f vooraf laten gaan door het morfisme r , gegeven door



$$r: \mathcal{C} \rightarrow \mathcal{C}, \quad (x, y) \mapsto (x, iy),$$

waarbij $i = \sqrt{-1}$. We kunnen r een aantal keer toepassen, dit geven we aan met r^k , waarbij $k = 0, 1, 2, 3$. Inderdaad geldt dat $f(r^k(x, y)) = f(x, i^k y) = x$. Het “spiegelen in de x -as” wordt nu gegeven door de afbeelding r^2 .

We definiëren een *Galois groep* G , waarin we alle symmetrieën van deze afbeelding $f: \mathcal{C} \rightarrow Y$ verzamelen. In dit geval is dus $G = \{\text{id}, r, r^2, r^3\}$, waarbij $\text{id} = r^0$ de afbeelding is die niets doet. We zeggen dat de groep G werkt op de kromme $\mathcal{C}$ via bovenstaande afbeeldingen.

Bij de kromme $\mathcal{C}$ kunnen we spreken over *reguliere differentiaalvormen*, waarbij men uitdrukkingen kan voorstellen van de vorm $\omega_1 = \frac{x dx}{y^3}$ en $\omega_2 = \frac{dx}{y^2}$. De groep G werkt op de ruimte van zulke differentiaalvormen, volgens dezelfde bovenstaande afbeeldingen, dus r stuurt ω_1 naar $r(\omega_1) = \frac{x dx}{(iy)^3} = i \frac{x dx}{y^3} = i\omega_1$ en op dezelfde manier stuurt r de vorm ω_2 naar $-\omega_2$.

We zien hier iets opmerkelijks gebeuren. We moeten r vier keer toepassen voordat ω_1 weer naar zichzelf gestuurd wordt, want $\omega_1 \xrightarrow{r} i\omega_1 \xrightarrow{r} -\omega_1 \xrightarrow{r} -i\omega_1 \xrightarrow{r} \omega_1$. Maar als we r toepassen op ω_2 , dan zien we dat we al na twee keer klaar zijn: $r^2(\omega_2) = \omega_2$. Grofweg kunnen we stellen dat de groep G anders werkt op ω_1 dan op ω_2 . We zeggen dat de *ruimte van reguliere differentiaalvormen* verschillende *representaties van de groep* G bevat.

Analoog aan het voorbeeld, behandelen we in deze scriptie afbeeldingen van krommen $f: X \rightarrow Y$ met een Galois groep G en vragen we ons af welke representaties van G voorkomen in de ruimte van reguliere differentiaalvormen. We werken nu niet meer met krommen over de complexe getallen $\mathbb{C}$, maar over een algemener lichaam k .